Nkambe bakery and frozen dessert business plan

Production recipes, product economics and a two-site operating plan

Prepared for the bakery owner • 3 October 2026

Revised: preparation ahead for meat, fish and chicken; product-specific reheating, oven methods and four optional recipes added in section 5A.

| Scope | Included |
|---|---|
| Core range | 20 products with complete shared formulas and assembly cards |
| Added savoury range | Four optional recipes: roast chicken burger, fish burger, fish wrap and beef mini pizza; see section 5A |
| Frozen range | 8 products and sale formats, with two popsicle size examples |
| Equipment | Diesel 16-tray oven, 40 × 60 cm trays, 20/15 kW inverters and 30 kWh battery storage at each site |
| Commercial basis | Illustrative FCFA inputs, computed yields and contribution margins |
| Status | Trial formulations and operational recommendations; supplier quotations and machine model validation outstanding |

Start with a small, repeatable menu. Prove taste, portion cost, safe handling and repeat purchase before filling the oven or expanding the range.

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# 1 Executive recommendation

Build the opening business around six dependable core products: family bread, condensed milk buns, sugar doughnuts, puff puff, fish rolls and beef meat pies. Add chicken shawarma only when its service station and raw-meat workflow pass a live trial. Start the frozen counter with one approved soft-serve base and small milk pops; add the larger pops only after customers accept the price and freezer throughput is measured.

This is a commercial hypothesis, not a forecast of Nkambe demand. No current Nkambe customer survey, supplier quotations, competitor-weight audit, footfall count or sales history was supplied. The menu prices

below are test points, not observed local market prices. Product margins use the same explicit illustrative ingredient-price list throughout.

There is a documented local sweet potato processing and bakery initiative. CLGF reported the Nkambe project in September 2025; that supports exploring a local supplier and product story, but does not establish today's supply, consumer demand or the performance of this report's 15% sweet potato formula. CLGF project report

| Rank | Opening priority | Decision condition |
|---|---|---|
| 1 | P01 Family bread | Blind comparison against locally purchased loaves at matched weight |
| 2 | P02 Milk buns | Target customers accept price after tasting |
| 3 | P04 Sugar doughnuts | Two consecutive batches meet oil, yield and texture checks |
| 4 | P06 Fish rolls | Chilled filling, cooking and transport logs pass |
| 5 | P07 Beef pies | Filling portion and cost remain controlled |
| 6 | P05 Puff puff | Sell in small waves; labour and oil do not erase contribution |
| 7 | P13 Chicken shawarma | Only after service drill and safe cooking verification |
| 8 | F01 Soft serve or F05 milk pop | Choose one first, based on manufacturer approval and cold-chain drill |

Make P03 sweet potato bread a small paid test, not the opening volume anchor. Keep pizza, beef shawarma, burgers, fried-chicken meals and dairy cake for later phases; they add value but also create different queues, allergens and temperature-control demands. The full 20-product library is a development range, not a recommended opening-day menu.

# 2 Equipment facts and operating assumptions

| Item | Owner-confirmed information | Still needed |
|---|---|---|
| Rotary oven | 16 trays; 40 × 60 cm; diesel heating; reported 3 kW electrical | Model, diesel litres/hour, maximum temperature, rack load, clearances and actual bake uniformity |
| Production power | Reported 20 kW inverter; 30 kWh battery storage | Continuous AC rating, phase distribution, surge rating, usable battery fraction, discharge current and PV panel kWp |
| Sales power | Reported 15 kW inverter; 30 kWh battery storage | Same details; input charging limits are not necessarily output limits |
| Mixing/forming | Spiral 6.5 kW; moulder 1 kW; mixer 2 kW; professional mixing pot | Minimum/maximum dough mass; pot heating/agitation specification |
| Heat appliances | Proofers 1 kW; fryers 5 kW; cookers 5 kW; shawarma 3.5 kW | Number of units per rating, electrical phase, fryer litres and recovery |
| Ice cream | Two machines: 4 kW and 1.5 kW; described as soft serve and gravity | Model and feed method of each; approved mixes, hopper cooling, minimum charge and cleaning cycle |
| Popsicles | Two mould sizes; one is twice the other | Working fill volumes, cavities, machine power, cycle time and brine specification |
| Cold chain | Owner reports substantial refrigeration/freezing at both sites | Measured pull-down capacity, temperature logs, space, connected load and backup runtime |

| Item | Owner-confirmed information | Still needed |
|---|---|---|
| Other | Bread slicer available | Slicer power, blender/hot plate availability, scales, probes, insulated transport equipment |

“Gravity” and “soft serve” are not mutually exclusive machine categories. A manufacturer’s catalogue explicitly lists gravity-fed soft-serve machines. The plan uses one conditional soft-serve trial base; it does not infer that the second machine is a batch freezer. Obtain model plates before choosing settings or loading homemade mix. Carpigiani soft serve categories

The new 30 kWh battery figure is recorded as nominal energy at each site. Solar array capacity remains unknown. The oven’s diesel supplies the heat: count both its electrical auxiliaries and diesel consumption. The electrical rating alone is not its full energy cost.

## Batch and tray capacity

A 40 × 60 cm tray has 0.24 $m^2$ of surface; sixteen trays have 3.84 $m^2$ before spacing and usable-area losses. That does not specify how many tins fit, the permissible rack weight, dough-mixer capacity or hourly output. Draw a tray layout with the actual tins. A trial of 12 milk buns per tray would mean 192 buns per rack, but it is an illustration, not a rated capacity. Do not fill all sixteen trays until airflow and bake uniformity have been checked.

The recipe library gives pilot quantities and a x5 planning production quantity. No mixer bowl or minimum machine charge has been verified. Use a smaller mixer or hand pilot if the spiral cannot mix the pilot load safely. Production quantities are for purchasing and scale-up trials, not permission to overload equipment. The professional mixing pot is not assumed to be a validated pasteurizer.

## Weights and formula conventions

All formulation quantities are grams, including liquids, to avoid density errors. Water alone may be approximated as 1 g/ml. Milk, oil, condensed milk and fruit puree must be weighed. An egg is provisionally 50 g edible contents; use the weighed mass, not a fixed count. Vegetable prices are per peeled/trimmed edible kilogram; meat is boneless purchase weight; canned fish is drained weight. Convert supplier gross-pack prices before use.

Yields are trial estimates after stated losses. Cost cooked components by saleable output. Frying oil entries cover usage/disposal, not the initial fryer fill; budget the initial fill separately as working stock.

# 3 The 20 core products

All selling ranges below are illustrative FCFA test prices. Service times assume ingredients and equipment are ready; they are not production lead times. Cold-chain levels refer to ingredients as well as the finished product. Shared-component identifiers lead to the complete formulas in section 4 or section 5.

| Product and customer | Role and portion | Price range | Equipment and service |
|---|---|---|---|
| P01 Soft family bread — Households; breakfast and evening | Daily traffic. 3 x 550 g dough; about 470–495 g baked each | 600–750 | Spiral, moulder, proofer, rotary oven; slicer after cooling. Moderate; counter service under 1 minute |

| Product and customer | Role and portion | Price range | Equipment and service |
|---|---|---|---|
| P02 Condensed milk breakfast buns — School/work breakfast and tea | Daily traffic. 20 x 90 g dough; about 78–82 g baked | 250–350 | Spiral, proofer, rotary oven. Moderate; under 1 minute |
| P03 Nkambe sweet potato milk loaf — Households and gifts | Signature. 3 x 580 g dough; about 495–522 g baked | 750–900 | Spiral, moulder, proofer, oven. Moderate; under 1 minute |
| P04 Sugar ring doughnuts — Students, commuters, tea breaks | Daily traffic. 25 x 70 g dough; about 66–73 g finished with sugar | 200–300 | Spiral, proofer, fryer. Moderate; under 1 minute |
| P05 Puff puff — School breaks and small cash purchases | Entry traffic. 60 x 30 g raw portions; about 27–31 g fried | 50–100 | Mixing bowl/pot, fryer. Low/moderate; under 1 minute |
| P06 Cameroon style fish rolls — Snack and light lunch | Daily traffic. 24 x 65 g dough + 25 g filling; about 82–88 g cooked | 300–400 | Mixer, cooker, fryer. Moderate; under 1 minute when safely held |
| P07 Beef meat pies — Lunch/snack for workers and households | Higher value. 22 x 75 g pastry + 50 g filling; about 112–118 g baked | 500–650 | Mixer, cooker, oven. Moderate; under 1 minute |

| Product and customer | Role and portion | Price range | Equipment and service |
|---|---|---|---|
| P08 Sausage rolls — Breakfast or savoury snack | Complementary. 24 x 65 g pastry + 40 g raw filling; about 88–96 g baked | 450–600 | Mixer, oven. Moderate; under 1 minute |
| P09 Creamy chicken pies — Lunch and premium snack | Higher value. 21 x 80 g pastry + 60 g filling; about 125–133 g baked | 650–800 | Mixer, cooker, oven. Moderate; under 1 minute |
| P10 Whole egg dough rolls — Breakfast and student lunch | Filling snack. 16 x 100 g dough + 50 g peeled cooked egg; about 140–148 g fried | 500–650 | Mixer, cooker, fryer. Moderate; under 1 minute |
| P11 Banana condensed milk cake slices — Tea, breakfast and household treats | Complementary. 3 x 1,000 g batter loaves; 24 slices about 105–113 g baked | 350–500 | Mixer, oven. Low/moderate; under 1 minute |
| P12 Personal chicken pizza — Individual lunch and evening treat | Higher value. 11 x 150 g dough + 160 g toppings; about 275–290 g baked | 1600–2200 | Mixer, oven, chilled toppings. Moderate; 8–12 minutes from raw base |
| P13 Complete chicken shawarma — Lunch and evening; workers and youth | Flagship. 1 wrap + 110 g cooked chicken + 145 g additions; about 345 g | 1800–2300 | Shawarma machine, hot plate, fryer, chilled rail. High; 3–5 minutes with prepared station |
| P14 Complete beef shawarma — Evening premium meal | Higher value. 1 wrap + 110 g cooked beef + 145 g additions; about 345 g | 2200–2700 | Shawarma/hot plate, fryer, chilled rail. High; 3–5 minutes |

| Product and customer | Role and portion | Price range | Equipment and service |
|---|---|---|---|
| P15 Crispy chicken burger — Individual lunch and evening | Higher value. 87 g bun + 110 g cooked coated chicken + 75 g garnish; about 272 g | 1700–2200 | Oven buns, fryer, hot plate, chilled rail. High; 4–8 minutes |
| P16 Beef cheeseburger — Lunch and evening | Higher value. 87 g bun + 75 g cooked patty + 105 g cheese/garnish; about 267 g | 1800–2400 | Oven buns, hot plate, chilled rail. Moderate/high; 6–9 minutes |

| Product and customer | Role and portion | Price range | Equipment and service |
|---|---|---|---|
| P17 Crispy chicken and fries box — Lunch and evening take-away | Meal. 220 g cooked coated chicken + 150 g fries + 30 g sauce | 2400–3000 | Fryer, cold raw station, hot holding. High; 6–10 minutes |
| P18 Chicken vegetable samosas — School snack, events and catering | Complementary. 50 x 30 g wrapper + 40 g filling + 2 g paste; about 65–72 g cooked | 250–400 | Mixer, cooker, fryer. High assembly labour; under 1 minute from safe holding |
| P19 Loaded chicken shawarma fries — Afternoon sharing snack or light meal | Complementary. 180 g fries + 80 g chicken + 25 g garlic sauce + 5 g hot sauce + 15 g pickle | 1700–2200 | Fryer, cooked chicken station, cold sauces. Moderate; 3–5 minutes |
| P20 Condensed milk soaked cake portions — Celebrations and chilled dessert purchases | Premium dessert. 950 g baked sponge + 800 g soak; 12 portions about 140 g after trim | 600–800 | Mixer, oven, refrigerator. Moderate; under 1 minute |

# Operational comparison

| Product | Shared inputs | Storage burden | Reason and principal risk |
|---|---|---|---|
| P01 Soft family bread | B01 | Low for unfilled day-of-sale bread | Repeat staple purchase. Risk: Local price and weight competition |
| P02 Condensed milk breakfast buns | B02, Egg | Low when plain; no fresh glaze | Premium aroma from a shared enriched dough. Risk: Dairy cost and sugar level |
| P03 Nkambe sweet potato milk loaf | B03 | Low for same-day plain loaf | Local identity and a testable differentiator. Risk: Flour consistency and smaller loaf volume |
| P04 Sugar ring doughnuts | B04, Oil, Sugar | Low for unfilled same-day service | Repeat snack with visible fresh batches. Risk: Oil control and unsold stock |
| P05 Puff puff | B05, Oil | Low for same-day plain sale | Affordable trial purchase and multipacks. Risk: Labour, oil use and low value per transaction |
| P06 Cameroon style fish rolls | B06, B08, Oil | High; fish filling and finished rolls | Distinct savoury snack. Risk: Fish price and temperature abuse |
| P07 Beef meat pies | B07, B09, Egg | High; cooked meat | Filling snack with shared pastry. Risk: Meat inflation and leakage |
| P08 Sausage rolls | B07, B10, Egg | High; raw meat preparation | Uses pie pastry and familiar format. Risk: Unclear supplier meat/allergen specification |
| P09 Creamy chicken pies | B07, B11, Egg | High; poultry and dairy | Creamy alternative to beef pie. Risk: Higher ingredient cost and cooling burden |
| P10 Whole egg dough rolls | B12, B13, Oil | High; cooked egg | Recognizable portion with a whole egg. Risk: Egg cost and undercooked thick dough |
| P11 Banana condensed milk cake slices | B14 | Moist cake; same-day policy | Uses sound ripe bananas and offers a non-fried snack. Risk: Variable banana moisture and unsold slices |
| P12 Personal chicken pizza | B15, B16, B18, Mozzarella, Onion trimmed, Pepper trimmed | High; poultry and cheese | Uses oven and existing cooked chicken. Risk: Cheese cost and oven bottleneck |

| Product | Shared inputs | Storage burden | Reason and principal risk |
|---|---|---|---|
| P13 Complete chicken shawarma | B17, B18, B24, B22, B23, B20, B21 | High; raw poultry, sauces and slaw | Complete portable meal. Risk: Meat yield, service speed and safe cone size |
| P14 Complete beef shawarma | B17, B19, B24, B22, B23, B35, B21 | High; meat, sesame sauce and slaw | Distinct beef and sesame flavour. Risk: Beef tenderness, cost and sesame allergen |
| P15 Crispy chicken burger | B25, B26, B28, Lettuce trimmed, Tomato trimmed, B23 | High; poultry and cold garnish | House bun and crisp chicken texture. Risk: Fryer capacity and raw-contact control |
| P16 Beef cheeseburger | B25, B27, Cheese slices, B28, Lettuce trimmed, Tomato trimmed, Onion trimmed, B23 | High; mince and garnish | House bun and a clearly measured beef portion. Risk: Beef cost and unsafe undercooking |
| P17 Crispy chicken and fries box | B26, B24, B28 | High; poultry and sauce | Larger meal using shared chicken preparation. Risk: Oil use, fryer queue and portion creep |
| P18 Chicken vegetable samosas | B29, B31, B30, Oil | High; chicken filling | Batch catering potential and shared ingredients. Risk: Labour and leaking seams |
| P19 Loaded chicken shawarma fries | B24, B18, B20, B21, B23 | High; cooked poultry and sauce | Raises basket value with existing safe components. Risk: Soggy fries and misuse of leftovers |
| P20 Condensed milk soaked cake portions | B32, B33 | High; soaked dairy cake | Distinct chilled treat using condensed milk. Risk: Cold chain and excess soak |

# 4 Core recipes and production standards

Use each assembly card together with its named base formulas. A base amount on a card is usable prepared mass, not necessarily a whole base batch. Scaling factor = required usable mass / base usable yield. The ingredient table then scales every raw ingredient by that factor. This prevents the common error of giving a filling batch that cannot fill the stated number of pieces.

All oven temperatures and process times below are starting trials for the diesel rotary oven, not manufacturer settings or recipe validation. Run a small load first, weigh cooled output and correct the process sheet. Use fresh instant dry yeast; confirm flour consistency and water temperature each day.

## S1 Food handling controls used throughout

Use a cleaned, calibrated probe. For this plan, adopt a single conservative house target of at least 75°C at the thickest centre for 30 seconds when cooking meat, poultry, fish and their fillings, and when reheating cooked portions. Check actual food temperature, not the oven dial. This updates the earlier separate meat limits for simpler staff operation; frozen dairy mix processing retains its separate F0 instructions. These are external benchmarks and a house choice, not verified Cameroon legal limits. UK food business cooking guidance

Keep chilled perishable ingredients at or below 5°C. For cooked food requiring cooling, use shallow pans, smaller portions or an ice bath: cool from 57°C to 21°C within 2 hours and to 5°C within 6 hours total. Log start, intermediate and final temperatures; a deep pot in a refrigerator does not prove compliance. These thresholds follow the cited FDA model guidance. FDA Food Code 2022 sections 3-501.14–16

Adopt a house hot-holding minimum of 63°C, checked at the food rather than the cabinet dial. This follows UK business guidance and is a separate house benchmark; it is not a statement of Cameroon law. Food with an unknown time/temperature history is not rescued by reheating. UK food business cooking guidance

House pilot policy: refrigerate components promptly, make only same-day quantities, and discard unsold high-risk prepared food at close unless a separately documented, validated storage plan applies. The same-day policy does not make a temperature-abused product safe. Quick pickles and cooked sauces remain refrigerated; their acidity has not been validated for ambient storage.

## Condensed milk formulation

Sweetened condensed milk supplies added sugar, milk solids and water. It is different from unsweetened evaporated milk and dry milk powder. For development calculations only, assume 45% added sucrose, 27% water and 28% milk solids including about 8% fat; these fractions sum to 100%. Naturally occurring lactose is inside the milk-solids fraction, so do not add total label sugars a second time.

Check the purchased label. One manufacturer example reports 55.9 g total sugars and 8.1 g fat per 100 g; total sugars are not the same as added sucrose. No brand specification is assumed to apply to your local tin. Nestlé example product specification

In B02, 150 g condensed milk adds approximately 67.5 g sucrose, 40.5 g water and 42 g milk solids, in addition to the separately listed 80 g sugar and 430 g water. The trial substitution on P02 approximates those functions, not identical flavour. For cost: 150 g condensed milk at 3,000 FCFA/kg costs 450 FCFA; 68 g sugar + 42 g whole milk powder + 40 g water costs about 244 FCFA, saving roughly 206 FCFA per B02 pilot, before any quality/yield change. Test both blind.

Use condensed milk selectively in buns, banana cake, soaked cake and frozen dairy products. Keep P01 bread, plain puff puff and most savouries free of it in the baseline to protect entry-level cost. Added dairy glaze changes the holding requirements even if the underlying bread is normally sold at ambient temperature.

## Complete base formulas

| Code | Base | Usable pilot yield g |
|---|---|---|
| B01 | Family bread dough | 1,700 |
| B02 | Condensed milk bun dough | 1,870 |
| B03 | Sweet potato milk bread dough | 1,790 |
| B04 | Yeast doughnut dough | 1,780 |
| B05 | Puff puff batter | 1,830 |
| B06 | Fish roll dough | 1,660 |
| B07 | Shortcrust pie pastry | 1,740 |
| B08 | Fish filling | 700 |
| B09 | Beef pie filling | 1,300 |
| B10 | Sausage roll filling | 980 |

| Code | Base | Usable pilot yield g |
|---|---|---|
| B11 | Creamy chicken filling | 1,300 |
| B12 | Egg roll dough | 1,650 |
| B13 | Hard boiled peeled eggs | 970 |
| B14 | Banana condensed milk cake batter | 3,010 |
| B15 | Pizza dough | 1,690 |
| B16 | Pizza tomato sauce | 750 |
| B17 | Shawarma flatbread | 1,500 |
| B18 | Cooked chicken shawarma meat | 750 |
| B19 | Cooked beef shawarma meat | 700 |
| B20 | Garlic mayonnaise sauce | 975 |
| B21 | Cooked hot sauce | 700 |
| B22 | Shawarma slaw | 975 |
| B23 | Refrigerator cucumber pickles | 650 |
| B24 | Crisp fries | 650 |
| B25 | Burger buns | 1,580 |
| B26 | Cooked crispy chicken pieces | 1,150 |
| B27 | Cooked beef burger patties | 750 |
| B28 | Burger sauce | 975 |
| B29 | Samosa wrapper dough | 1,510 |
| B30 | Samosa sealing paste | 98 |
| B31 | Chicken vegetable samosa filling | 1,000 |
| B32 | Milk cake sponge batter | 950 |
| B33 | Three milk soak | 990 |
| B34 | Condensed milk glaze | 335 |
| B35 | Beef shawarma sesame sauce | 975 |

# B01 Family bread dough

Pilot input 1,715 g; target usable output 1,700 g. Planning production batch is x5: input 8,575 g and usable output 8,500 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Bread flour | 1,000 | 5,000 | 100.00% |

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Water | 600 | 3,000 | 60.00% |
| Sugar | 50 | 250 | 5.00% |
| Margarine | 35 | 175 | 3.50% |
| Salt | 18 | 90 | 1.80% |
| Yeast | 12 | 60 | 1.20% |

Baker's percentages use 1,000 g total flour as 100%; added milk/egg/fat are weighed ingredients, not equivalent to water.

1. Weigh ingredients. Reserve 30 g of the listed water; add only if the flour absorbs it. The stated yield assumes all 600 g is used.
2. Mix flour, water, sugar and yeast about 3 minutes at low speed. Add salt, then mix 4–7 minutes at the mixer's dough-development speed. Add margarine during the final 2 minutes. Stop at a thin elastic window and a dough temperature near 26–28°C.
3. Rest covered 15 minutes. Divide, pre-round, rest 10 minutes, then mould and tin as specified on the product card.
4. Trial proof at 30–35°C with enough humidity to prevent skinning, approximately 45–75 minutes. Bake when a gentle indentation returns slowly; do not use the clock alone.

Yield and process note: Input 1,715 g; allow 15 g bowl/handling loss. No improver assumed. If one is used, follow its label and re-cost.

Illustrative ingredient cost: 835 FCFA per pilot batch; 491.3 FCFA/kg usable output. This excludes packaging and operating costs.

# B02 Condensed milk bun dough

Pilot input 1,895 g; target usable output 1,870 g. Planning production batch is x5: input 9,475 g and usable output 9,350 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Bread flour | 1,000 | 5,000 | 100.00% |
| Water | 430 | 2,150 | 43.00% |
| Condensed milk | 150 | 750 | 15.00% |
| Sugar | 80 | 400 | 8.00% |
| Egg | 100 | 500 | 10.00% |
| Margarine | 80 | 400 | 8.00% |
| Whole milk powder | 20 | 100 | 2.00% |
| Salt | 16 | 80 | 1.60% |

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Yeast | 15 | 75 | 1.50% |
| Vanilla | 4 | 20 | 0.40% |

Baker's percentages use 1,000 g total flour as 100%; added milk/egg/fat are weighed ingredients, not equivalent to water.

1. Dissolve condensed milk in water. Mix with flour, sugar, milk powder, yeast and egg on low for 3 minutes; reserve 25 g water until dough consistency is clear.
2. Add salt, develop 4–6 minutes, then add margarine gradually and mix another 2–4 minutes to an elastic film. Target 26–28°C dough; stop before overheating.
3. Rest covered 15 minutes. Portion and round with a tight surface; use the product card for bun weights.
4. Proof around 30–35°C for approximately 50–80 minutes until light and a gentle indentation returns slowly.

Yield and process note: Input 1,895 g; 25 g handling allowance. Egg wash is listed separately on the product card.

Illustrative ingredient cost: 1,765 FCFA per pilot batch; 943.6 FCFA/kg usable output. This excludes packaging and operating costs.

## B03 Sweet potato milk bread dough

Pilot input 1,812 g; target usable output 1,790 g. Planning production batch is x5: input 9,060 g and usable output 8,950 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Bread flour | 850 | 4,250 | 85.00% |
| Sweet potato flour | 150 | 750 | 15.00% |
| Water | 570 | 2,850 | 57.00% |
| Condensed milk | 100 | 500 | 10.00% |
| Sugar | 40 | 200 | 4.00% |
| Margarine | 50 | 250 | 5.00% |
| Whole milk powder | 20 | 100 | 2.00% |
| Salt | 18 | 90 | 1.80% |
| Yeast | 14 | 70 | 1.40% |

Baker's percentages use 1,000 g total flour as 100%; added milk/egg/fat are weighed ingredients, not equivalent to water.

1. Blend the two flours thoroughly; baker's percentages use their combined 1,000 g. Reserve 40 g water for absorption adjustment.

2. Mix all except margarine for 3 minutes slowly, then develop 5–8 minutes. Add margarine and mix until extensible. Expect a less strong window than wheat-only bread; excessive mixing will not restore absent gluten.
3. Rest 15 minutes, divide and mould gently. Proof around 30–33°C to a resilient loaf, stopping before the surface weakens.
4. Use the product card for baking. Record actual loaf volume, sweetness and day-of-sale softness before increasing sweet potato flour.

Yield and process note: Input 1,812 g; 22 g handling allowance. Start at 15% sweet potato flour, not an untested high replacement ratio.

Illustrative ingredient cost: 1,309 FCFA per pilot batch; 731.0 FCFA/kg usable output. This excludes packaging and operating costs.

## B04 Yeast doughnut dough

Pilot input 1,808 g; target usable output 1,780 g. Planning production batch is x5: input 9,040 g and usable output 8,900 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Bread flour | 1,000 | 5,000 | 100.00% |
| Water | 440 | 2,200 | 44.00% |
| Egg | 100 | 500 | 10.00% |
| Sugar | 120 | 600 | 12.00% |
| Margarine | 80 | 400 | 8.00% |
| Whole milk powder | 30 | 150 | 3.00% |
| Salt | 15 | 75 | 1.50% |
| Yeast | 18 | 90 | 1.80% |
| Nutmeg | 2 | 10 | 0.20% |
| Vanilla | 3 | 15 | 0.30% |

Baker's percentages use 1,000 g total flour as 100%; added milk/egg/fat are weighed ingredients, not equivalent to water.

1. Mix flour, water, egg, sugar, milk powder and yeast 3 minutes slowly. Add salt and mix 4–6 minutes; finish by incorporating margarine, nutmeg and vanilla.
2. Stop at elastic dough near 26–28°C. Rest covered 15 minutes; weigh individual pieces instead of repeatedly rerolling scraps.
3. Round or form rings; proof around 30°C for approximately 40–65 minutes until light, not fragile. Uncover only briefly to make transfer manageable.

4. Fry as specified on the product card; use a thermometer and allow oil temperature to recover between loads.

Yield and process note: Input 1,808 g; 28 g handling allowance. No filling is included.

Illustrative ingredient cost: 1,417 FCFA per pilot batch; 796.2 FCFA/kg usable output. This excludes packaging and operating costs.

## B05 Puff puff batter

Pilot input 1,846 g; target usable output 1,830 g. Planning production batch is x5: input 9,230 g and usable output 9,150 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Plain flour | 1,000 | 5,000 | 100.00% |
| Water | 700 | 3,500 | 70.00% |
| Sugar | 120 | 600 | 12.00% |
| Salt | 12 | 60 | 1.20% |
| Yeast | 12 | 60 | 1.20% |
| Nutmeg | 2 | 10 | 0.20% |

Baker's percentages use 1,000 g total flour as 100%; added milk/egg/fat are weighed ingredients, not equivalent to water.

1. Mix dry ingredients, then add water and beat 4–6 minutes to a cohesive sticky batter without dry flour.
2. Cover and ferment near 26–30°C for approximately 45–75 minutes, until bubbly and substantially expanded. Stir once gently if bubbles are very uneven.
3. Use a calibrated scoop to deliver 30 g portions. Trial fry one piece before loading the fryer.
4. Fry at 165–175°C, turning for even colour and checking the centre is set, typically 4–6 minutes depending on size.

Yield and process note: Input 1,846 g; 16 g bowl loss. Fermentation time is conditional on temperature.

Illustrative ingredient cost: 846 FCFA per pilot batch; 462.5 FCFA/kg usable output. This excludes packaging and operating costs.

## B06 Fish roll dough

Pilot input 1,679 g; target usable output 1,660 g. Planning production batch is x5: input 8,395 g and usable output 8,300 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Plain flour | 1,000 | 5,000 | 100.00% |
| Water | 470 | 2,350 | 47.00% |

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Margarine | 100 | 500 | 10.00% |
| Sugar | 60 | 300 | 6.00% |
| Whole milk powder | 20 | 100 | 2.00% |
| Salt | 15 | 75 | 1.50% |
| Yeast | 12 | 60 | 1.20% |
| Nutmeg | 2 | 10 | 0.20% |

Baker's percentages use 1,000 g total flour as 100%; added milk/egg/fat are weighed ingredients, not equivalent to water.

1. Mix flour, sugar, powder, yeast and water on low for 3 minutes. Add salt, margarine and nutmeg; mix another 4–6 minutes until smooth, with moderate gluten development.
2. Rest covered 20 minutes. Divide and roll as specified, without making the wrapper excessively thick.
3. Use fully chilled filling; seal clean dough edges firmly, with no filling trapped in the seam.
4. Rest assembled rolls 15–25 minutes before frying. They should be slightly aerated rather than fully doubled.

Yield and process note: Input 1,679 g; 19 g handling loss.

Illustrative ingredient cost: 1,068 FCFA per pilot batch; 643.6 FCFA/kg usable output. This excludes packaging and operating costs.

# B07 Shortcrust pie pastry

Pilot input 1,773 g; target usable output 1,740 g. Planning production batch is x5: input 8,865 g and usable output 8,700 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Plain flour | 1,000 | 5,000 | 100.00% |
| Margarine | 400 | 2,000 | 40.00% |
| Water | 290 | 1,450 | 29.00% |
| Egg | 60 | 300 | 6.00% |
| Salt | 15 | 75 | 1.50% |
| Baking powder | 8 | 40 | 0.80% |

Baker's percentages use 1,000 g total flour as 100%; added milk/egg/fat are weighed ingredients, not equivalent to water.

1. Keep fat and water cool. Mix flour, salt and baking powder; rub in margarine to coarse crumbs.

2. Add egg and most of the water. Fold just until cohesive; add the remainder only if needed. Do not develop a bread-like gluten network.
3. Cover and refrigerate 30–45 minutes. Roll to about 3 mm on a lightly floured surface.
4. Weigh cut portions, use chilled filling and seal. Rework clean trimmings only once; include any additional scrap in the recorded yield loss.

Yield and process note: Input 1,773 g; 33 g planned trimming/bowl loss after limited rework. Dusting flour must be recorded if material.

Illustrative ingredient cost: 1,541 FCFA per pilot batch; 885.9 FCFA/kg usable output. This excludes packaging and operating costs.

# B08 Fish filling

Pilot input 783 g; target usable output 700 g. Planning production batch is x5: input 3,915 g and usable output 3,500 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Canned fish drained | 500 | 2,500 | — |
| Onion trimmed | 150 | 750 | — |
| Carrot trimmed | 80 | 400 | — |
| Oil | 25 | 125 | — |
| Garlic peeled | 10 | 50 | — |
| Ginger peeled | 8 | 40 | — |
| Salt | 5 | 25 | — |
| Black pepper | 2 | 10 | — |
| Paprika | 3 | 15 | — |

1. Check fish for bones. Weigh drained edible fish; the cost input is per drained kilogram, not unopened-can weight.
2. Soften finely chopped onion and carrot in oil 5–7 minutes. Add garlic, ginger and spices, then fish.
3. Cook while stirring until cohesive and fairly dry. Weigh the finished filling; target 700 g, with no free liquid.
4. Spread in shallow pans and actively cool under control S1. Chill before filling dough; never leave a large warm pot on a counter.

Yield and process note: Input 783 g; target 700 g after evaporation. If yield differs, recompute cost per gram.

Illustrative ingredient cost: 2,322 FCFA per pilot batch; 3,317.1 FCFA/kg usable output. This excludes packaging and operating costs.

# B09 Beef pie filling

Pilot input 1,506 g; target usable output 1,300 g. Planning production batch is x5: input 7,530 g and usable output 6,500 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Beef mince | 500 | 2,500 | — |
| Potato peeled | 300 | 1,500 | — |
| Carrot trimmed | 150 | 750 | — |
| Onion trimmed | 200 | 1,000 | — |
| Oil | 30 | 150 | — |
| Plain flour | 35 | 175 | — |
| Water | 250 | 1,250 | — |
| Garlic peeled | 12 | 60 | — |
| Ginger peeled | 8 | 40 | — |
| Salt | 12 | 60 | — |
| Black pepper | 3 | 15 | — |
| Thyme | 2 | 10 | — |
| Curry powder | 4 | 20 | — |

1. Dice vegetables to about 5 mm. Brown beef in oil, breaking lumps; cook onion, garlic and ginger with it.
2. Add potato, carrot, seasonings and water. Simmer until vegetables soften; stir in flour dispersed in a little reserved water.
3. Cook until thick without raw flour taste and verify ground-meat control S1. Target 1,300 g finished filling; reduce gently if wet.
4. Rapidly cool in shallow pans under S1, cover after cooling and refrigerate. Use cold in pastry.

Yield and process note: Input 1,506 g; 206 g expected evaporation. Do not top up with cold untreated liquid after cooking.

Illustrative ingredient cost: 3,015 FCFA per pilot batch; 2,319.1 FCFA/kg usable output. This excludes packaging and operating costs.

# B10 Sausage roll filling

Pilot input 1,000 g; target usable output 980 g. Planning production batch is x5: input 5,000 g and usable output 4,900 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Sausage meat | 750 | 3,750 | — |

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Onion trimmed | 100 | 500 | — |
| Breadcrumbs | 70 | 350 | — |
| Egg | 60 | 300 | — |
| Garlic peeled | 10 | 50 | — |
| Black pepper | 2 | 10 | — |
| Paprika | 3 | 15 | — |
| Salt | 5 | 25 | — |

1. Use chilled food-grade sausage meat; verify species and allergens with supplier. Reduce the added salt if the supplied meat is already strongly seasoned.
2. Finely mince onion and garlic, then combine all ingredients just until evenly distributed.
3. Keep covered and cold; portion 40 g cylinders for assembly. Keep raw-meat tools away from cooked products.
4. Cook fully inside pastry according to S1; oven colour alone does not establish meat safety.

Yield and process note: Input 1,000 g; 20 g handling loss. Yield is raw filling; cooking loss is applied in the product card.

Illustrative ingredient cost: 3,356 FCFA per pilot batch; 3,424.5 FCFA/kg usable output. This excludes packaging and operating costs.

# B11 Creamy chicken filling

Pilot input 1,502 g; target usable output 1,300 g. Planning production batch is x5: input 7,510 g and usable output 6,500 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Chicken boneless | 600 | 3,000 | — |
| Potato peeled | 150 | 750 | — |
| Carrot trimmed | 100 | 500 | — |
| Onion trimmed | 120 | 600 | — |
| Pepper trimmed | 80 | 400 | — |
| Margarine | 30 | 150 | — |
| Plain flour | 45 | 225 | — |
| UHT milk | 250 | 1,250 | — |
| Water | 100 | 500 | — |
| Garlic peeled | 10 | 50 | — |

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Salt | 12 | 60 | — |
| Black pepper | 3 | 15 | — |
| Thyme | 2 | 10 | — |

1. Dice raw boneless chicken separately from vegetables. Cook chicken thoroughly under S1, then chop with cleaned utensils.
2. Soften onion, potato, carrot and pepper in margarine. Add garlic and flour; cook briefly, then gradually stir in milk and water.
3. Add cooked chicken and seasonings; simmer to a thick spoonable filling. Verify final centre temperature under S1; target 1,300 g.
4. Cool actively in shallow pans under S1 and refrigerate. This is a refrigerated filling, not a room-temperature display food.

Yield and process note: Input 1,502 g; target 1,300 g cooked. Raw chicken cost and cooking loss are included.

Illustrative ingredient cost: 3,283 FCFA per pilot batch; 2,525.0 FCFA/kg usable output. This excludes packaging and operating costs.

# B12 Egg roll dough

Pilot input 1,674 g; target usable output 1,650 g. Planning production batch is x5: input 8,370 g and usable output 8,250 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Plain flour | 1,000 | 5,000 | 100.00% |
| Water | 470 | 2,350 | 47.00% |
| Sugar | 80 | 400 | 8.00% |
| Margarine | 70 | 350 | 7.00% |
| Whole milk powder | 25 | 125 | 2.50% |
| Salt | 15 | 75 | 1.50% |
| Yeast | 12 | 60 | 1.20% |
| Nutmeg | 2 | 10 | 0.20% |

Baker’s percentages use 1,000 g total flour as 100%; added milk/egg/fat are weighed ingredients, not equivalent to water.

1. Mix flour, water, sugar, powder and yeast 3 minutes slowly. Add salt and fat; mix another 5–7 minutes until smooth.
2. Rest covered 15 minutes. Weigh 100 g pieces and flatten evenly.
3. Enclose a fully cooked, chilled, peeled egg; eliminate air pockets and close the seam completely.

4. Proof assembled rolls 20–35 minutes until slightly expanded, then fry according to the product card.

Yield and process note: Input 1,674 g; 24 g handling loss. Hard-boiled egg is a separate component.

Illustrative ingredient cost: 1,053 FCFA per pilot batch; 638.1 FCFA/kg usable output. This excludes packaging and operating costs.

# B13 Hard boiled peeled eggs

Pilot input 1,000 g; target usable output 970 g. Planning production batch is x5: input 5,000 g and usable output 4,850 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Egg | 1,000 | 5,000 | — |

1. The formula uses edible egg weight. Purchase intact eggs, keep shell-handling separate, and wash hands after handling.
2. Lower into boiling water, simmer until yolks and whites are fully set; a starting trial is 10–12 minutes for medium eggs, then verify by cutting a sample.
3. Cool rapidly in potable ice water, peel hygienically and refrigerate. Weigh peeled cooked yield.
4. Portion by actual egg size; the model uses 50 g peeled eggs. Do not store peeled eggs at ambient temperature.

Yield and process note: 970 g expected cooked edible yield from 1,000 g raw edible equivalent. Shell cost is already converted into the egg price input.

Illustrative ingredient cost: 2,200 FCFA per pilot batch; 2,268.0 FCFA/kg usable output. This excludes packaging and operating costs.

# B14 Banana condensed milk cake batter

Pilot input 3,061 g; target usable output 3,010 g. Planning production batch is x5: input 15,305 g and usable output 15,050 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Plain flour | 1,000 | 5,000 | — |
| Banana peeled | 900 | 4,500 | — |
| Egg | 400 | 2,000 | — |
| Oil | 300 | 1,500 | — |
| Sugar | 220 | 1,100 | — |
| Condensed milk | 200 | 1,000 | — |
| Baking powder | 20 | 100 | — |
| Baking soda | 5 | 25 | — |

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Salt | 8 | 40 | — |
| Vanilla | 5 | 25 | — |
| Cinnamon | 3 | 15 | — |

1. Mash sound ripe peeled bananas. Discard mouldy or fermented fruit; ripeness is not permission to use spoilage.
2. Whisk egg, sugar, condensed milk and oil, then banana. Sift flour, raising agents, salt and cinnamon together.
3. Fold dry into wet just until no dry flour remains. Deposit promptly into lined tins.
4. Bake under the product card. Do not leave mixed batter waiting while another oven load finishes.

Yield and process note: Input 3,061 g; 51 g bowl loss. No peanut is included in the baseline.

Illustrative ingredient cost: 3,686 FCFA per pilot batch; 1,224.5 FCFA/kg usable output. This excludes packaging and operating costs.

# B15 Pizza dough

Pilot input 1,708 g; target usable output 1,690 g. Planning production batch is x5: input 8,540 g and usable output 8,450 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Bread flour | 1,000 | 5,000 | 100.00% |
| Water | 620 | 3,100 | 62.00% |
| Salt | 20 | 100 | 2.00% |
| Sugar | 20 | 100 | 2.00% |
| Yeast | 8 | 40 | 0.80% |
| Oil | 40 | 200 | 4.00% |

Baker’s percentages use 1,000 g total flour as 100%; added milk/egg/fat are weighed ingredients, not equivalent to water.

1. Mix flour, water, sugar and yeast 3 minutes slowly; add salt and develop 5–7 minutes. Add oil near the end.
2. Cover and ferment approximately 60–90 minutes at 24–27°C until aerated. Fold once if needed.
3. Divide into 150 g balls, rest covered 20–30 minutes, then stretch to 18–20 cm without flattening every bubble at the rim.
4. Top immediately before baking; use cooked chicken and chilled sauce, not raw poultry.

Yield and process note: Input 1,708 g; 18 g handling loss.

Illustrative ingredient cost: 789 FCFA per pilot batch; 466.9 FCFA/kg usable output. This excludes packaging and operating costs.

## B16 Pizza tomato sauce

Pilot input 977 g; target usable output 750 g. Planning production batch is x5: input 4,885 g and usable output 3,750 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Tomato trimmed | 800 | 4,000 | — |
| Onion trimmed | 100 | 500 | — |
| Oil | 30 | 150 | — |
| Garlic peeled | 15 | 75 | — |
| Sugar | 15 | 75 | — |
| Salt | 12 | 60 | — |
| Oregano | 3 | 15 | — |
| Black pepper | 2 | 10 | — |

1. Chop tomatoes and onions; soften onion and garlic in oil.
2. Add tomatoes, sugar and seasoning; simmer and stir until thick enough not to flood pizza dough.
3. Blend with sanitized equipment if a smooth sauce is wanted; finish at 750 g target yield.
4. Rapidly cool and refrigerate under S1. Portion 40 g per pizza.

Yield and process note: Input 977 g; evaporation target 227 g. Tomato ripeness changes yield.

Illustrative ingredient cost: 884 FCFA per pilot batch; 1,178.4 FCFA/kg usable output. This excludes packaging and operating costs.

## B25 Burger buns

Pilot input 1,802 g; target usable output 1,580 g. Planning production batch is x5: input 9,010 g and usable output 7,900 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Bread flour | 1,000 | 5,000 | 100.00% |
| Water | 540 | 2,700 | 54.00% |
| Egg | 80 | 400 | 8.00% |
| Sugar | 65 | 325 | 6.50% |
| Margarine | 60 | 300 | 6.00% |
| Whole milk powder | 25 | 125 | 2.50% |

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Salt | 18 | 90 | 1.80% |
| Yeast | 14 | 70 | 1.40% |

Baker's percentages use 1,000 g total flour as 100%; added milk/egg/fat are weighed ingredients, not equivalent to water.

1. Mix flour, water, egg, sugar, powder and yeast 3 minutes slowly; add salt and develop 5–7 minutes. Add margarine at the end.
2. Rest 15 minutes; divide into 18 pieces of 99 g before normal handling loss. Round, rest and flatten gently.
3. Proof around 30–35°C until light. Bake at a trial 185–195°C for 14–18 minutes; cool on racks.
4. Target 1,580 g baked buns, about eighteen 87 g buns. For costing the output below is corrected to cooked usable mass.

Yield and process note: Input 1,802 g. Cooked output 1,580 g includes handling and baking losses; no egg wash or sesame included.

Illustrative ingredient cost: 1,192 FCFA per pilot batch; 754.7 FCFA/kg usable output. This excludes packaging and operating costs.

# B26 Cooked crispy chicken pieces

Pilot input 1,772 g; target usable output 1,150 g. Planning production batch is x5: input 8,860 g and usable output 5,750 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Chicken boneless | 1,000 | 5,000 | — |
| Yogurt plain | 150 | 750 | — |
| Water | 100 | 500 | — |
| Salt | 16 | 80 | — |
| Garlic peeled | 15 | 75 | — |
| Ginger peeled | 10 | 50 | — |
| Paprika | 8 | 40 | — |
| Black pepper | 3 | 15 | — |
| Plain flour | 280 | 1,400 | — |
| Corn starch | 70 | 350 | — |
| Oil | 120 | 600 | — |

1. Cut chicken into even boneless portions. Marinate refrigerated with yogurt, water, 10 g salt, garlic and ginger for 4–8 hours.

2. Mix flour, starch, remaining salt, paprika and pepper. Coat drained chicken, dip only into a separate working portion of marinade, then coat again. Discard all raw-contact surplus.
3. Fry in small loads at a trial 170–175°C, roughly 5–8 minutes for flattened pieces, until the thickest part meets S1 poultry temperature.
4. Drain on a rack; weigh the cooked coated output. Target 1,150 g saleable product including coating. Service is immediate or controlled hot holding.

Yield and process note: Gross input includes excess coating, discarded marinade and 120 g frying-oil allowance. All are charged; 1,150 g output is a trial yield, not an absorption guarantee.

Illustrative ingredient cost: 4,963 FCFA per pilot batch; 4,315.6 FCFA/kg usable output. This excludes packaging and operating costs.

## B27 Cooked beef burger patties

Preparation-ahead option: section 5A gives the full oven, cooling, portioning and reheating method. Its final-yield adjustments apply when this base is cooked, chilled and reheated.

Pilot input 1,028 g; target usable output 750 g. Planning production batch is x5: input 5,140 g and usable output 3,750 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Beef mince | 1,000 | 5,000 | — |
| Salt | 12 | 60 | — |
| Black pepper | 3 | 15 | — |
| Paprika | 3 | 15 | — |
| Oil | 10 | 50 | — |

1. Use mince with supplier-confirmed fat content, provisionally 15–20%. Keep chilled; do not add water or breadcrumbs.
2. Mix seasoning briefly, portion ten raw patties of approximately 102 g and make a shallow centre depression.
3. Cook on a lightly oiled hot plate, turning as needed, until the centre reaches the S1 ground-beef limit. Probe from the side into the thickest centre.
4. Target ten 75 g cooked patties; weigh actual yield. Assemble immediately or hold under S1.

Yield and process note: Input 1,028 g; cooked yield 750 g. Oil includes pan-use allowance.

Illustrative ingredient cost: 4,572 FCFA per pilot batch; 6,095.7 FCFA/kg usable output. This excludes packaging and operating costs.

## B28 Burger sauce

Pilot input 983 g; target usable output 975 g. Planning production batch is x5: input 4,915 g and usable output 4,875 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Mayonnaise pasteurized | 650 | 3,250 | — |
| Ketchup | 220 | 1,100 | — |
| Mustard | 50 | 250 | — |
| B23 Refrigerator cucumber pickles | 60 | 300 | — |
| Paprika | 3 | 15 | — |

1. Finely chop drained B23 pickles. Combine with pasteurized mayonnaise, ketchup, mustard and paprika.
2. Mix with sanitized utensils and portion into small clean service containers.
3. Label and refrigerate under S1; replace containers instead of topping up.
4. Use 25 g per burger and record remaining sauce at service close.

Yield and process note: Input 983 g; 8 g bowl loss. Mustard, egg and other supplier-declared allergens may apply.

Illustrative ingredient cost: 2,281 FCFA per pilot batch; 2,339.6 FCFA/kg usable output. This excludes packaging and operating costs.

## B29 Samosa wrapper dough

Pilot input 1,538 g; target usable output 1,510 g. Planning production batch is x5: input 7,690 g and usable output 7,550 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Plain flour | 1,000 | 5,000 | 100.00% |
| Water | 460 | 2,300 | 46.00% |
| Oil | 60 | 300 | 6.00% |
| Salt | 18 | 90 | 1.80% |

Baker's percentages use 1,000 g total flour as 100%; added milk/egg/fat are weighed ingredients, not equivalent to water.

1. Mix flour and salt with oil, then add water to make a firm smooth dough. Knead 5–7 minutes.
2. Cover and rest 30 minutes. Divide 30 g portions and roll thinly into circles; cut/fold into cones as preferred.
3. Make sealing paste from B30; fill with 40 g chilled B31 and exclude trapped air.
4. Seal all seams completely and fry promptly in small batches as specified on the product card.

Yield and process note: Input 1,538 g; 28 g handling/trimming loss. No yeast.

Illustrative ingredient cost: 756 FCFA per pilot batch; 500.3 FCFA/kg usable output. This excludes packaging and operating costs.

# B30 Samosa sealing paste

Pilot input 100 g; target usable output 98 g. Planning production batch is x5: input 500 g and usable output 490 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Plain flour | 40 | 200 | — |
| Water | 60 | 300 | — |

1. Whisk water into flour to a thick smooth paste.
2. Prepare a small quantity for one assembly session; keep covered and cool.
3. Apply 2 g to each seam; cook the sealed pastry thoroughly.
4. Discard the working remainder after assembly; never use it as a ready-to-eat sauce.

Yield and process note: Input 100 g; 2 g bowl loss.

Illustrative ingredient cost: 26 FCFA per pilot batch; 268.4 FCFA/kg usable output. This excludes packaging and operating costs.

# B31 Chicken vegetable samosa filling

Pilot input 1,141 g; target usable output 1,000 g. Planning production batch is x5: input 5,705 g and usable output 5,000 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Chicken boneless | 350 | 1,750 | — |
| Potato peeled | 350 | 1,750 | — |
| Carrot trimmed | 120 | 600 | — |
| Onion trimmed | 150 | 750 | — |
| Oil | 30 | 150 | — |
| Water | 100 | 500 | — |
| Garlic peeled | 10 | 50 | — |
| Ginger peeled | 10 | 50 | — |
| Salt | 12 | 60 | — |
| Curry powder | 6 | 30 | — |
| Cumin | 3 | 15 | — |

1. Cook finely diced chicken thoroughly under S1 using separate raw-meat utensils.
2. Soften onion, garlic and ginger in oil; add diced potato, carrot and water, then simmer until tender.
3. Fold in chicken and seasonings; cook to a dry filling and target 1,000 g cooked mass.

4. Rapidly cool in shallow trays under S1; use cold at 40 g per samosa.

Yield and process note: Input 1,141 g; target 1,000 g after cooking.

Illustrative ingredient cost: 2,106 FCFA per pilot batch; 2,106.3 FCFA/kg usable output. This excludes packaging and operating costs.

## B32 Milk cake sponge batter

Pilot input 1,105 g; target usable output 950 g. Planning production batch is x5: input 5,525 g and usable output 4,750 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Plain flour | 300 | 1,500 | — |
| Egg | 450 | 2,250 | — |
| Sugar | 220 | 1,100 | — |
| UHT milk | 80 | 400 | — |
| Oil | 40 | 200 | — |
| Baking powder | 8 | 40 | — |
| Salt | 3 | 15 | — |
| Vanilla | 4 | 20 | — |

1. Whip eggs and sugar in the mixing machine 6–10 minutes until thick and ribboning; use a clean bowl.
2. Sift flour, baking powder and salt. Fold into the foam gently, then fold in milk, oil and vanilla pre-mixed with a little batter.
3. Deposit into a lined shallow pan. Bake at a trial 170–180°C for 22–30 minutes until springy with no wet batter on a skewer.
4. Cool the sponge on a rack, then weigh. Target 950 g baked sponge; do not soak warm cake and leave it unrefrigerated.

Yield and process note: Input 1,105 g; about 25 g bowl and 130 g bake loss. Output used for costing is 950 g baked sponge.

Illustrative ingredient cost: 1,602 FCFA per pilot batch; 1,686.5 FCFA/kg usable output. This excludes packaging and operating costs.

## B33 Three milk soak

Pilot input 1,000 g; target usable output 990 g. Planning production batch is x5: input 5,000 g and usable output 4,950 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Condensed milk | 400 | 2,000 | — |

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| UHT milk | 400 | 2,000 | — |
| Cream 35% | 200 | 1,000 | — |

1. Mix cold pasteurized/UHT dairy in a sanitized jug.
2. Keep refrigerated; weigh the amount applied to each fully cooled sponge.
3. Use 800 g soak per 950 g B32 sponge and allow absorption under refrigeration.
4. Do not reuse liquid drained from a cake or serving tray.

Yield and process note: Input 1,000 g; 10 g handling loss. A full base leaves 190 g when one full B32 sponge is soaked.

Illustrative ingredient cost: 2,980 FCFA per pilot batch; 3,010.1 FCFA/kg usable output. This excludes packaging and operating costs.

## B34 Condensed milk glaze

Pilot input 342 g; target usable output 335 g. Planning production batch is x5: input 1,710 g and usable output 1,675 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Condensed milk | 200 | 1,000 | — |
| Icing sugar | 100 | 500 | — |
| Butter | 20 | 100 | — |
| Water | 20 | 100 | — |
| Vanilla | 2 | 10 | — |

1. Melt butter gently. Stir into condensed milk with sifted icing sugar and vanilla.
2. Add potable water gradually to the measured consistency; use all listed water for the model yield.
3. Apply a thin measured layer to products; prepare only the quantity needed for the session.
4. Treat the diluted glaze and glazed goods under refrigerated control S1 unless a competent assessment validates another holding method.

Yield and process note: Input 342 g; 7 g bowl loss. No ambient shelf-life claim.

Illustrative ingredient cost: 904 FCFA per pilot batch; 2,698.8 FCFA/kg usable output. This excludes packaging and operating costs.

# Finished product assembly cards

## P01 Soft family bread

Quality target: Fine even crumb, thin golden crust, clean slices and a soft but fully baked centre.

Pilot: 3 saleable units. Planning production batch: 15 units. Portion: 3 x 550 g dough; about 470–495 g baked each. Scale all components x5; validate capacity and losses before adoption.

| Prepared component or ingredient | Pilot g | Production g |
|---|---|---|
| B01 Family bread dough | 1,700 | 8,500 |

1. Make B01. Weigh three 550 g dough pieces; the 50 g surplus covers QC/test dough and is charged to these loaves.
2. Pre-round, rest 10 minutes and mould into greased tins. Proof as B01; leave room for oven spring rather than overproofing to the tin edge.
3. Trial bake at 195–205°C for 25–32 minutes. Check a representative loaf centre near 94–96°C as a baking-quality cue, not a universal safety limit.
4. De-tin immediately. Cool on racks until no warmth/condensation remains before slicing and bagging. Verify cooled sale weight and slicing consistency.

Holding and packaging: A — Unfilled bread/plain fried snack: quality policy is sale on production day after protected cooling. No multi-day ambient safety claim. Keep clean, dry and protected; record discard or separately approved reuse. Adding dairy glaze/filling changes the class.

Troubleshooting: Dense crumb: insufficient proof/development; collapsed top: excessive proof or underbake; gummy slices: sliced too warm.

Substitution: Use the strongest consistent local wheat flour; re-trial water rather than assuming equal absorption.

Release check: weigh ingredients and 3 portions; log actual usable yield and rejects; compare centre, colour and texture to the target; confirm required temperatures and allergen label; record packaging and unsold stock.

| Batch ingredient cost | Saleable units | Ingredients/unit | Pack/unit | Other variable/unit | Test price | Contribution/unit |
|---|---|---|---|---|---|---|
| 835 | 3 | 278.4 | 30 | 55 | 650 | 286.6 |

All values in FCFA. Contribution margin 44.1%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 230.9 FCFA. At 80% sell-through it falls to 195.8 FCFA per sold unit under the section 7 model.

## P02 Condensed milk breakfast buns

Quality target: Rounded golden buns, shreddable soft crumb, clear milk aroma without a wet surface.

Pilot: 20 saleable units. Planning production batch: 100 units. Portion: 20 x 90 g dough; about 78–82 g baked. Scale all components x5; validate capacity and losses before adoption.

| Prepared component or ingredient | Pilot g | Production g |
|---|---|---|
| B02 Condensed milk bun dough | 1,870 | 9,350 |
| Egg | 40 | 200 |

1. Make B02 and portion twenty 90 g buns. Charge the remaining 70 g dough to QC/variation.
2. Round tightly, place with expansion space, proof using B02, then brush lightly with the 40 g weighed egg wash; discard unused raw-contact wash.
3. Bake at a trial 180–190°C for 14–18 minutes. Rotate only if safe and the oven's distribution test indicates a need.
4. Cool completely and bag. Start with plain buns; an optional 8 g B34 glaze adds refrigerated handling and additional cost, so it is excluded from baseline pricing.

Holding and packaging: A — Unfilled bread/plain fried snack: quality policy is sale on production day after protected cooling. No multi-day ambient safety claim. Keep clean, dry and protected; record discard or separately approved reuse. Adding dairy glaze/filling changes the class.

Troubleshooting: Tight crumb: add fat too early or underproof; very dark crust: excessive sugar or heat; wrinkling: underbaking.

Substitution: Budget trial: replace each 150 g condensed milk with 68 g sugar, 42 g whole milk powder and 40 g water; sensory and cost recheck required.

Release check: weigh ingredients and 20 portions; log actual usable yield and rejects; compare centre, colour and texture to the target; confirm required temperatures and allergen label; record packaging and unsold stock.

| Batch ingredient cost | Saleable units | Ingredients/unit | Pack/unit | Other variable/unit | Test price | Contribution/unit |
|---|---|---|---|---|---|---|
| 1,853 | 20 | 92.6 | 10 | 20 | 250 | 127.4 |

All values in FCFA. Contribution margin 51.0%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 108.8 FCFA. At 80% sell-through it falls to 96.7 FCFA per sold unit under the section 7 model.

# P03 Nkambe sweet potato milk loaf

Quality target: Warm yellow/brown crumb depending on flour, mild sweetness and enough elasticity for clean slices.

Pilot: 3 saleable units. Planning production batch: 15 units. Portion: 3 x 580 g dough; about 495–522 g baked. Scale all components x5; validate capacity and losses before adoption.

| Prepared component or ingredient | Pilot g | Production g |
|---|---|---|
| B03 Sweet potato milk bread dough | 1,790 | 8,950 |

1. Make B03 using sound food-grade sweet potato flour. Divide three 580 g loaves; charge 50 g remaining dough to QC.
2. Mould gently, tin and proof to resilience; do not force the height of a wheat-only loaf.

3. Trial bake at 190–200°C for 27–33 minutes; verify fully baked centre and record cooled mass.
4. Cool and bag with an accurate ingredient description. Test this against P01 at two prices before committing a large daily volume.

Holding and packaging: A — Unfilled bread/plain fried snack: quality policy is sale on production day after protected cooling. No multi-day ambient safety claim. Keep clean, dry and protected; record discard or separately approved reuse. Adding dairy glaze/filling changes the class.

Troubleshooting: Low volume: weak flour or too much sweet potato; gummy interior: underbake/high water; gritty texture: coarse flour.

Substitution: If sweet potato flour is unavailable, sell P01 under its own name; cooked sweet potato puree is not a direct gram-for-gram substitute.

Release check: weigh ingredients and 3 portions; log actual usable yield and rejects; compare centre, colour and texture to the target; confirm required temperatures and allergen label; record packaging and unsold stock.

| Batch ingredient cost | Saleable units | Ingredients/unit | Pack/unit | Other variable/unit | Test price | Contribution/unit |
|---|---|---|---|---|---|---|
| 1,309 | 3 | 436.2 | 35 | 60 | 800 | 268.8 |

All values in FCFA. Contribution margin 33.6%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 181.6 FCFA. At 80% sell-through it falls to 136.0 FCFA per sold unit under the section 7 model.

# P04 Sugar ring doughnuts

Quality target: Light crumb, an even ring, thin sugar finish and no raw centre or greasy bite.

Pilot: 25 saleable units. Planning production batch: 125 units. Portion: 25 x 70 g dough; about 66–73 g finished with sugar. Scale all components x5; validate capacity and losses before adoption.

| Prepared component or ingredient | Pilot g | Production g |
|---|---|---|
| B04 Yeast doughnut dough | 1,780 | 8,900 |
| Oil | 200 | 1,000 |
| Sugar | 120 | 600 |

1. Prepare B04, portion twenty-five 70 g rings; 30 g spare dough is charged to QC.
2. Proof until light. Fry at 170–175°C about 60–100 seconds per side as a trial, adjusting to centre doneness and colour.
3. Drain on a rack; apply 120 g sugar across the batch. Oil entry is usage/disposal allocation, not 200 g guaranteed absorption.
4. Sell in small replenishment waves. Glazed variant: replace sugar dusting with 8 g B34 per unit and use refrigerated handling; do not hide its extra cost in this baseline.

Holding and packaging: A — Unfilled bread/plain fried snack: quality policy is sale on production day after protected cooling. No multi-day ambient safety claim. Keep clean, dry and protected; record discard or separately approved reuse. Adding dairy glaze/filling changes the class.

Troubleshooting: Dark/raw centre: oil too hot; oily heavy crumb: underproof or low oil recovery; collapse: overproof.

Substitution: Omit milk powder only after water/texture trials; do not substitute baking powder for yeast in this formula.

Release check: weigh ingredients and 25 portions; log actual usable yield and rejects; compare centre, colour and texture to the target; confirm required temperatures and allergen label; record packaging and unsold stock.

| Batch ingredient cost | Saleable units | Ingredients/unit | Pack/unit | Other variable/unit | Test price | Contribution/unit |
|---|---|---|---|---|---|---|
| 1,833 | 25 | 73.3 | 10 | 25 | 250 | 141.7 |

All values in FCFA. Contribution margin 56.7%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 127.0 FCFA. At 80% sell-through it falls to 114.6 FCFA per sold unit under the section 7 model.

# P05 Puff puff

Quality target: Rounded pieces with an aerated cooked centre, even colour and no sour overfermentation.

Pilot: 60 saleable units. Planning production batch: 300 units. Portion: 60 x 30 g raw portions; about 27–31 g fried. Scale all components x5; validate capacity and losses before adoption.

| Prepared component or ingredient | Pilot g | Production g |
|---|---|---|
| B05 Puff puff batter | 1,830 | 9,150 |
| Oil | 180 | 900 |

1. Make B05. Portion sixty 30 g scoops; the remaining 30 g is charged to trial pieces.
2. Fry at 165–175°C in small batches, turning until uniformly golden and the centre is fully set.
3. Drain and count; put five in a ventilated paper pack only when ordered. Do not seal steaming puff puff in plastic.
4. At the example price, a five-piece pack is 350–375 FCFA only if the discounted contribution remains acceptable.

Holding and packaging: A — Unfilled bread/plain fried snack: quality policy is sale on production day after protected cooling. No multi-day ambient safety claim. Keep clean, dry and protected; record discard or separately approved reuse. Adding dairy glaze/filling changes the class.

Troubleshooting: Flat pieces: thin or overfermented batter; hard centre: low hydration/underproof; oily finish: low oil recovery.

Substitution: Use a 25 g raw portion for a lower-price test only after recalculating yield and clearly preserving value.

Release check: weigh ingredients and 60 portions; log actual usable yield and rejects; compare centre, colour and texture to the target; confirm required temperatures and allergen label; record packaging and unsold stock.

| Batch ingredient cost | Saleable units | Ingredients/unit | Pack/unit | Other variable/unit | Test price | Contribution/unit |
|---|---|---|---|---|---|---|
| 1,134 | 60 | 18.9 | 3 | 8 | 75 | 45.1 |

All values in FCFA. Contribution margin 60.1%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 41.3 FCFA. At 80% sell-through it falls to 37.6 FCFA per sold unit under the section 7 model.

## P06 Cameroon style fish rolls

Quality target: Crisp golden exterior, rolled seam and a clearly present dry fish filling.

Pilot: 24 saleable units. Planning production batch: 120 units. Portion: 24 x 65 g dough + 25 g filling; about 82–88 g cooked. Scale all components x5; validate capacity and losses before adoption.

| Prepared component or ingredient | Pilot g | Production g |
|---|---|---|
| B06 Fish roll dough | 1,660 | 8,300 |
| B08 Fish filling | 600 | 3,000 |
| Oil | 180 | 900 |

1. Make B06 and B08 separately. Use 1,560 g dough and 600 g cold filling for 24 rolls; 100 g dough is QC allowance. A full 700 g B08 batch leaves 100 g allocated separately.
2. Roll 65 g dough pieces, add 25 g filling, close edges and seam, then rest briefly as B06.
3. Fry at a trial 165–175°C until golden and filling is thoroughly reheated under S1. Check the thickest sample.
4. Drain and use holding class H. Transport finished rolls hot under control; an alternative is chilled assembled uncooked rolls for final frying at the sales site under section 5A. Do not chill finished rolls for a second reheating of the fish filling.

Holding and packaging: H — The final bake/fry reheats the previously cooked chilled filling or topping. Serve promptly or hold hot under S1, with a house quality target of at most 2 hours. Do not cool finished leftovers for another reheat. Follow the product-specific transport and finishing route in section 5A.

Troubleshooting: Burst seams: wet filling or poor seal; gummy spiral: wrapper too thick; fishy flavour: unsuitable fish/storage.

Substitution: Use cooked deboned fresh fish only with measured purchase-to-edible yield and revised cost.

Release check: weigh ingredients and 24 portions; log actual usable yield and rejects; compare centre, colour and texture to the target; confirm required temperatures and allergen label; record packaging and unsold stock.

| Batch ingredient cost | Saleable units | Ingredients/unit | Pack/unit | Other variable/unit | Test price | Contribution/unit |
|---|---|---|---|---|---|---|
| 3,347 | 24 | 139.4 | 10 | 30 | 300 | 120.6 |

All values in FCFA. Contribution margin 40.2%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 92.7 FCFA. At 80% sell-through it falls to 75.7 FCFA per sold unit under the section 7 model.

# P07 Beef meat pies

Quality target: Short flaky bite, full corners, intact seal and visible beef/vegetable filling.

Pilot: 22 saleable units. Planning production batch: 110 units. Portion: 22 x 75 g pastry + 50 g filling; about 112–118 g baked. Scale all components x5; validate capacity and losses before adoption.

| Prepared component or ingredient | Pilot g | Production g |
|---|---|---|
| B07 Shortcrust pie pastry | 1,740 | 8,700 |
| B09 Beef pie filling | 1,100 | 5,500 |
| Egg | 45 | 225 |

1. Prepare B07 and B09. Cut twenty-two 75 g pastry portions and add 50 g chilled filling each. Pastry surplus is 90 g; full B09 yields 200 g unused filling.
2. Seal with clean edges, vent, brush using 45 g allocated egg wash and discard raw-contact leftovers.
3. Trial bake at 190–200°C for 22–28 minutes. Check base colour, pastry doneness and filling temperature under S1.
4. Apply H holding and package without trapping excessive steam.

Holding and packaging: H — The final bake/fry reheats the previously cooked chilled filling or topping. Serve promptly or hold hot under S1, with a house quality target of at most 2 hours. Do not cool finished leftovers for another reheat. Follow the product-specific transport and finishing route in section 5A.

Troubleshooting: Tough pastry: overmixing; soggy base: wet/hot filling; leakage: overfilling or weak seam.

Substitution: Replace part of beef with potato only as a separately labelled economy formula with recalculated portions.

Release check: weigh ingredients and 22 portions; log actual usable yield and rejects; compare centre, colour and texture to the target; confirm required temperatures and allergen label; record packaging and unsold stock.

| Batch ingredient cost | Saleable units | Ingredients/unit | Pack/unit | Other variable/unit | Test price | Contribution/unit |
|---|---|---|---|---|---|---|
| 4,191 | 22 | 190.5 | 20 | 35 | 500 | 254.5 |

All values in FCFA. Contribution margin 50.9%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 216.4 FCFA. At 80% sell-through it falls to 193.1 FCFA per sold unit under the section 7 model.

# P08 Sausage rolls

Quality target: Crisp pastry with a fully cooked moist meat centre and clean slice.

Pilot: 24 saleable units. Planning production batch: 120 units. Portion: 24 x 65 g pastry + 40 g raw filling; about 88–96 g baked. Scale all components x5; validate capacity and losses before adoption.

| Prepared component or ingredient | Pilot g | Production g |
|---|---|---|
| B07 Shortcrust pie pastry | 1,740 | 8,700 |

| Prepared component or ingredient | Pilot g | Production g |
|---|---|---|
| B10 Sausage roll filling | 960 | 4,800 |
| Egg | 45 | 225 |

1. Make B07 and keep B10 cold. Portion 24 rolls with 65 g pastry and 40 g raw filling each. Pastry surplus is 180 g; B10 batch leaves 20 g.
2. Form cylinders, enclose and seal pastry, apply 45 g allocated egg wash and vent the tops.
3. Trial bake at 190–200°C for 23–30 minutes. Probe meat to S1 ground-meat limit, or poultry limit if sausage contains poultry.
4. Hold and transport under H. Record any grease leakage before adjusting meat fat or crumb content.

Holding and packaging: H — Serve immediately or hold hot under S1. With the specified raw sausage filling, a fully baked roll may instead be rapidly cooled and reheated once using section 5A. House hot-display quality target: at most 2 hours. Record the actual filling heat history.

Troubleshooting: Raw centre: rolls too thick or oven too hot; greasy pastry: high-fat filling; splitting: poor vent/seal.

Substitution: Supplier-seasoned mince can replace sausage meat only after checking salt, species and allergen details.

Release check: weigh ingredients and 24 portions; log actual usable yield and rejects; compare centre, colour and texture to the target; confirm required temperatures and allergen label; record packaging and unsold stock.

| Batch ingredient cost | Saleable units | Ingredients/unit | Pack/unit | Other variable/unit | Test price | Contribution/unit |
|---|---|---|---|---|---|---|
| 4,928 | 24 | 205.3 | 15 | 35 | 450 | 194.7 |

All values in FCFA. Contribution margin 43.3%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 153.6 FCFA. At 80% sell-through it falls to 130.8 FCFA per sold unit under the section 7 model.

# P09 Creamy chicken pies

Quality target: Golden short pastry, moist chicken pieces and a thick sauce that does not run.

Pilot: 21 saleable units. Planning production batch: 105 units. Portion: 21 x 80 g pastry + 60 g filling; about 125–133 g baked. Scale all components x5; validate capacity and losses before adoption.

| Prepared component or ingredient | Pilot g | Production g |
|---|---|---|
| B07 Shortcrust pie pastry | 1,740 | 8,700 |
| B11 Creamy chicken filling | 1,260 | 6,300 |
| Egg | 45 | 225 |

1. Make B07 and B11. Use twenty-one 80 g pastry portions and 60 g chilled filling each; dough spare 60 g, full B11 spare 40 g.

2. Seal, vent and apply allocated egg wash. Keep waiting trays refrigerated.
3. Trial bake at 190–200°C for 23–30 minutes; verify filling reaches S1 reheating temperature.
4. Use H holding; limit hot display batches because pastry quality declines before safety-controlled holding necessarily ends.

Holding and packaging: H — The final bake/fry reheats the previously cooked chilled filling or topping. Serve promptly or hold hot under S1, with a house quality target of at most 2 hours. Do not cool finished leftovers for another reheat. Follow the product-specific transport and finishing route in section 5A.

Troubleshooting: Soggy pastry: loose sauce; dry chicken: prolonged hot holding; split pies: overfilled.

Substitution: Use cooked thigh from a separately costed plain preparation; do not assume raw and cooked chicken weights are interchangeable.

Release check: weigh ingredients and 21 portions; log actual usable yield and rejects; compare centre, colour and texture to the target; confirm required temperatures and allergen label; record packaging and unsold stock.

| Batch ingredient cost | Saleable units | Ingredients/unit | Pack/unit | Other variable/unit | Test price | Contribution/unit |
|---|---|---|---|---|---|---|
| 4,822 | 21 | 229.6 | 20 | 40 | 650 | 360.4 |

All values in FCFA. Contribution margin 55.4%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 314.5 FCFA. At 80% sell-through it falls to 288.0 FCFA per sold unit under the section 7 model.

# P10 Whole egg dough rolls

Quality target: A whole fully cooked egg centred inside evenly fried dough, with no wet seam.

Pilot: 16 saleable units. Planning production batch: 80 units. Portion: 16 x 100 g dough + 50 g peeled cooked egg; about 140–148 g fried. Scale all components x5; validate capacity and losses before adoption.

| Prepared component or ingredient | Pilot g | Production g |
|---|---|---|
| B12 Egg roll dough | 1,650 | 8,250 |
| B13 Hard boiled peeled eggs | 800 | 4,000 |
| Oil | 180 | 900 |

1. Prepare B12 and B13. Use sixteen 100 g dough pieces and sixteen cooked eggs averaging 50 g each; charge 50 g dough surplus to QC.
2. Wrap eggs evenly, close seams tightly and proof 20–35 minutes.
3. Trial fry at 165–170°C for 7–10 minutes, turning. Split a sample and check both dough doneness and reheated centre under S1.
4. Use H holding. Full B13 output is 970 g: 170 g remains for separately recorded use or the next scaled batch, within the same safe production plan.

Holding and packaging: H — The final bake/fry reheats the previously cooked chilled filling or topping. Serve promptly or hold hot under S1, with a house quality target of at most 2 hours. Do not cool finished leftovers for

another reheat. Follow the product-specific transport and finishing route in section 5A.

Troubleshooting: Burnt outside/raw inner dough: too high oil temperature; bursting: trapped air; uneven portions: egg sizes not checked.

Substitution: Smaller eggs require a separately priced smaller product, not silent substitution.

Release check: weigh ingredients and 16 portions; log actual usable yield and rejects; compare centre, colour and texture to the target; confirm required temperatures and allergen label; record packaging and unsold stock.

| Batch ingredient cost | Saleable units | Ingredients/unit | Pack/unit | Other variable/unit | Test price | Contribution/unit |
|---|---|---|---|---|---|---|
| 3,155 | 16 | 197.2 | 15 | 35 | 500 | 252.8 |

All values in FCFA. Contribution margin 50.6%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 213.3 FCFA. At 80% sell-through it falls to 191.0 FCFA per sold unit under the section 7 model.

# P11 Banana condensed milk cake slices

Quality target: Moist but set crumb, clear banana aroma, clean slices and no compressed wet line.

Pilot: 24 saleable units. Planning production batch: 120 units. Portion: 3 x 1,000 g batter loaves; 24 slices about 105–113 g baked. Scale all components x5; validate capacity and losses before adoption.

| Prepared component or ingredient | Pilot g | Production g |
|---|---|---|
| B14 Banana condensed milk cake batter | 3,010 | 15,050 |

1. Prepare B14 and fill three suitable loaf tins with 1,000 g batter each; 10 g is bowl/QC allocation. Choose tins with enough expansion space.
2. Trial bake at 170–180°C for 45–60 minutes; use a clean skewer and spring-back rather than time alone.
3. Cool fully, weigh each loaf, and cut each into eight equal slices; expected bake loss 10–16%.
4. Package slices only after cooling. Use moist-cake holding class M; do not assert a multi-day ambient shelf life.

Holding and packaging: M — Moist banana cake: sell on production day; protect during short display and refrigerate reserve stock at or below 5°C. Ambient multi-day shelf life requires a competent water-activity/pH assessment and validation.

Troubleshooting: Sunken centre: underbake/excess raising; gummy crumb: wet bananas; dry slice: excess bake or thin portions.

Substitution: Swap some banana for pumpkin only through a new moisture trial; optional nuts require new allergen control and cost.

Release check: weigh ingredients and 24 portions; log actual usable yield and rejects; compare centre, colour and texture to the target; confirm required temperatures and allergen label; record packaging and unsold stock.

| Batch ingredient cost | Saleable units | Ingredients/unit | Pack/unit | Other variable/unit | Test price | Contribution/unit |
|---|---|---|---|---|---|---|
| 3,686 | 24 | 153.6 | 20 | 25 | 400 | 201.4 |

All values in FCFA. Contribution margin 50.4%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 170.7 FCFA. At 80% sell-through it falls to 151.8 FCFA per sold unit under the section 7 model.

# P12 Personal chicken pizza

Quality target: Crisp underside, cooked centre, melted cheese and a visible 50 g chicken portion.

Pilot: 11 saleable units. Planning production batch: 55 units. Portion: 11 x 150 g dough + 160 g toppings; about 275–290 g baked. Scale all components x5; validate capacity and losses before adoption.

| Prepared component or ingredient | Pilot g | Production g |
|---|---|---|
| B15 Pizza dough | 1,690 | 8,450 |
| B16 Pizza tomato sauce | 440 | 2,200 |
| B18 Cooked chicken shawarma meat | 550 | 2,750 |
| Mozzarella | 550 | 2,750 |
| Onion trimmed | 110 | 550 |
| Pepper trimmed | 110 | 550 |

1. Make B15. Use eleven 150 g bases; 40 g dough spare is charged to QC.
2. Per base add 40 g B16, 50 g cooked B18, 50 g mozzarella, 10 g onion and 10 g pepper. Keep prepared toppings refrigerated.
3. Trial bake at 230–250°C for 7–11 minutes only if the diesel oven is rated for this temperature. Check base and reheated chicken under S1; hotter baking needs a separate oven schedule.
4. Serve immediately. If the rotary oven cannot deliver a suitable base at its approved setting, postpone pizza or use a tested par-baked base workflow with controlled cooling.

Holding and packaging: H — The final bake/fry reheats the previously cooked chilled filling or topping. Serve promptly or hold hot under S1, with a house quality target of at most 2 hours. Do not cool finished leftovers for another reheat. Follow the product-specific transport and finishing route in section 5A.

Troubleshooting: Wet centre: excess sauce or low oven recovery; burnt cheese: long bake; high cost: uncontrolled cheese handfuls.

Substitution: Use less expensive melting cheese only after taste/melt testing; never replace measured chicken with extra fries.

Release check: weigh ingredients and 11 portions; log actual usable yield and rejects; compare centre, colour and texture to the target; confirm required temperatures and allergen label; record packaging and unsold stock.

| Batch ingredient cost | Saleable units | Ingredients/unit | Pack/unit | Other variable/unit | Test price | Contribution/unit |
|---|---|---|---|---|---|---|
| 8,581 | 11 | 780.1 | 90 | 80 | 1800 | 849.9 |

All values in FCFA. Contribution margin 47.2%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 693.9 FCFA. At 80% sell-through it falls to 612.4 FCFA per sold unit under the section 7 model.

# P13 Complete chicken shawarma

Quality target: Flexible wrap, hot juicy meat, crisp fries and balanced sauce with a tight clean roll.

Pilot: 16 saleable units. Planning production batch: 80 units. Portion: 1 wrap + 110 g cooked chicken + 145 g additions; about 345 g. Scale all components x5; validate capacity and losses before adoption.

| Prepared component or ingredient | Pilot g | Production g |
|---|---|---|
| B17 Shawarma flatbread | 1,500 | 7,500 |
| B18 Cooked chicken shawarma meat | 1,760 | 8,800 |
| B24 Crisp fries | 800 | 4,000 |
| B22 Shawarma slaw | 720 | 3,600 |
| B23 Refrigerator cucumber pickles | 240 | 1,200 |
| B20 Garlic mayonnaise sauce | 480 | 2,400 |
| B21 Cooked hot sauce | 80 | 400 |

1. Prepare the exact components in section 5; this card costs sixteen full standard wraps. B17 delivers 16 x 90 g bread with 60 g allowance.
2. For each wrap weigh 110 g cooked B18, 50 g B24 fries, 45 g B22 slaw, 15 g B23 pickles, 30 g B20 garlic sauce and optional 5 g B21 hot sauce; hot sauce cost is included.
3. Assemble using separate hot and ready-to-eat utensils. Fold sides, roll firmly and toast about 30–60 seconds per side on the clean hot plate without crushing.
4. Wrap in food-contact paper and hand over immediately. Use fresh safe service stock; a filled wrap is not a long-hold product.

Holding and packaging: I — Assemble to order: keep hot and cold components separately controlled under S1 until assembly. Hand over promptly and recommend immediate consumption; do not build an ambient ready-wrapped stock.

Troubleshooting: Wet wrap: undrained slaw or excess sauce; dry bread: overbaking; chewy meat: wrong cut or excessive holding.

Substitution: Use thigh as default; breast needs separate moisture/yield tests. A garlic-only mild option omits hot sauce without changing meat.

Release check: weigh ingredients and 16 portions; log actual usable yield and rejects; compare centre, colour and texture to the target; confirm required temperatures and allergen label; record packaging and unsold stock.

| Batch ingredient cost | Saleable units | Ingredients/unit | Pack/unit | Other variable/unit | Test price | Contribution/unit |
|---|---|---|---|---|---|---|
| 15,321 | 16 | 957.5 | 80 | 60 | 2000 | 902.5 |

All values in FCFA. Contribution margin 45.1%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 711.0 FCFA. At 80% sell-through it falls to 628.1 FCFA per sold unit under the section 7 model.

# P14 Complete beef shawarma

Quality target: Tender sliced beef with a balanced sesame sauce, no gristle and an intact wrap.

Pilot: 16 saleable units. Planning production batch: 80 units. Portion: 1 wrap + 110 g cooked beef + 145 g additions; about 345 g. Scale all components x5; validate capacity and losses before adoption.

| Prepared component or ingredient | Pilot g | Production g |
|---|---|---|
| B17 Shawarma flatbread | 1,500 | 7,500 |
| B19 Cooked beef shawarma meat | 1,760 | 8,800 |
| B24 Crisp fries | 800 | 4,000 |
| B22 Shawarma slaw | 720 | 3,600 |
| B23 Refrigerator cucumber pickles | 240 | 1,200 |
| B35 Beef shawarma sesame sauce | 480 | 2,400 |
| B21 Cooked hot sauce | 80 | 400 |

1. Use the same sixteen-wrap bread allocation as P13, substituting cooked B19 beef and B35 sesame sauce.
2. Assemble each with 110 g cooked beef, 50 g fries, 45 g slaw, 15 g drained pickles, 30 g sesame sauce and optional 5 g hot sauce.
3. Toast and wrap as P13. Check the raw-purchase quantity using 70% beef yield before purchasing.
4. Serve immediately and disclose sesame. Keep sauce utensils separate from the garlic-only option.

Holding and packaging: I — Assemble to order: keep hot and cold components separately controlled under S1 until assembly. Hand over promptly and recommend immediate consumption; do not build an ambient ready-wrapped stock.

Troubleshooting: Tough beef: wrong cut or slicing direction; bitter sauce: poor tahini; wet wrap: excess liquid.

Substitution: B20 can replace B35 at 30 g per wrap for a sesame-free recipe only when cross-contact can also be controlled.

Release check: weigh ingredients and 16 portions; log actual usable yield and rejects; compare centre, colour and texture to the target; confirm required temperatures and allergen label; record packaging and unsold stock.

| Batch ingredient cost | Saleable units | Ingredients/unit | Pack/unit | Other variable/unit | Test price | Contribution/unit |
|---|---|---|---|---|---|---|
| 16,955 | 16 | 1,059.7 | 80 | 65 | 2400 | 1,195.3 |

All values in FCFA. Contribution margin 49.8%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 983.4 FCFA. At 80% sell-through it falls to 894.1 FCFA per sold unit under the section 7 model.

# P15 Crispy chicken burger

Quality target: Toasted soft bun, crisp fully cooked chicken and fresh measured garnish.

Pilot: 10 saleable units. Planning production batch: 50 units. Portion: 87 g bun + 110 g cooked coated chicken + 75 g garnish; about 272 g. Scale all components x5; validate capacity and losses before adoption.

| Prepared component or ingredient | Pilot g | Production g |
|---|---|---|
| B25 Burger buns | 870 | 4,350 |
| B26 Cooked crispy chicken pieces | 1,100 | 5,500 |
| B28 Burger sauce | 250 | 1,250 |
| Lettuce trimmed | 150 | 750 |
| Tomato trimmed | 250 | 1,250 |
| B23 Refrigerator cucumber pickles | 100 | 500 |

1. Make or allocate B25 buns; a full batch supplies eighteen buns, of which ten are used here.
2. Use ten 110 g cooked coated B26 portions. Toast split buns on a clean hot plate.
3. Per burger add 25 g B28 sauce, 15 g lettuce, 25 g tomato and 10 g drained pickles. Keep garnish cold until assembly.
4. Serve immediately in a vented box. Optional cheese is 20 g and costs extra; fries are not included.

Holding and packaging: I — Assemble to order: keep hot and cold components separately controlled under S1 until assembly. Hand over promptly and recommend immediate consumption; do not build an ambient ready-wrapped stock.

Troubleshooting: Soggy crust: covered hot chicken or early assembly; raw middle: chicken too thick; slippery burger: watery tomato.

Substitution: A grilled B18 chicken burger is a separate 110 g cooked-meat option with recalculated cost.

Release check: weigh ingredients and 10 portions; log actual usable yield and rejects; compare centre, colour and texture to the target; confirm required temperatures and allergen label; record packaging and unsold stock.

| Batch ingredient cost | Saleable units | Ingredients/unit | Pack/unit | Other variable/unit | Test price | Contribution/unit |
|---|---|---|---|---|---|---|
| 6,602 | 10 | 660.2 | 70 | 65 | 1800 | 1,004.8 |

All values in FCFA. Contribution margin 55.8%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 872.8 FCFA. At 80% sell-through it falls to 806.0 FCFA per sold unit under the section 7 model.

# P16 Beef cheeseburger

Quality target: Fully cooked juicy patty, melted cheese, toasted bun and a balanced bite.

Pilot: 10 saleable units. Planning production batch: 50 units. Portion: 87 g bun + 75 g cooked patty + 105 g cheese/garnish; about 267 g. Scale all components x5; validate capacity and losses before adoption.

| Prepared component or ingredient | Pilot g | Production g |
|---|---|---|
| B25 Burger buns | 870 | 4,350 |
| B27 Cooked beef burger patties | 750 | 3,750 |
| Cheese slices | 200 | 1,000 |
| B28 Burger sauce | 250 | 1,250 |
| Lettuce trimmed | 150 | 750 |
| Tomato trimmed | 250 | 1,250 |
| Onion trimmed | 100 | 500 |
| B23 Refrigerator cucumber pickles | 100 | 500 |

1. Prepare ten B27 patties from the full base batch, verifying S1 ground-meat temperature.
2. Toast ten B25 buns. Melt 20 g cheese on each cooked patty briefly.
3. Add 25 g sauce, 15 g lettuce, 25 g tomato, 10 g onion and 10 g pickles per burger.
4. Serve immediately. Advertise raw patty mass only if clearly labelled; the cooked target is 75 g, not 100 g.

Holding and packaging: I — Assemble to order: keep hot and cold components separately controlled under S1 until assembly. Hand over promptly and recommend immediate consumption; do not build an ambient ready-wrapped stock.

Troubleshooting: Dry patty: lean mince or overholding; domed patty: poor shaping; cold centre: heat too high/short cook.

Substitution: Do not add breadcrumbs to conceal a beef reduction; a lower-cost patty needs a disclosed new formula.

Release check: weigh ingredients and 10 portions; log actual usable yield and rejects; compare centre, colour and texture to the target; confirm required temperatures and allergen label; record packaging and unsold stock.

| Batch ingredient cost | Saleable units | Ingredients/unit | Pack/unit | Other variable/unit | Test price | Contribution/unit |
|---|---|---|---|---|---|---|
| 7,807 | 10 | 780.7 | 70 | 60 | 1900 | 989.3 |

All values in FCFA. Contribution margin 52.1%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 833.2 FCFA. At 80% sell-through it falls to 761.6 FCFA per sold unit under the section 7 model.

# P17 Crispy chicken and fries box

Quality target: Crisp chicken pieces, hot fries, sauce in a separate cup and no trapped steam.

Pilot: 5 saleable units. Planning production batch: 25 units. Portion: 220 g cooked coated chicken + 150 g fries + 30 g sauce. Scale all components x5; validate capacity and losses before adoption.

| Prepared component or ingredient | Pilot g | Production g |
|---|---|---|
| B26 Cooked crispy chicken pieces | 1,100 | 5,500 |
| B24 Crisp fries | 750 | 3,750 |
| B28 Burger sauce | 150 | 750 |

1. Prepare B26 as smaller even strips, applying the same full cooking checks. Allocate 220 g cooked coated chicken per box.
2. Finish 150 g B24 fries per box near handover; salt consistently.
3. Place 30 g B28 sauce in a separate cup. Use a vented food box.
4. Serve promptly. Do not describe this boneless-strip portion as a specified number of bone-in pieces.

Holding and packaging: I — Assemble to order: keep hot and cold components separately controlled under S1 until assembly. Hand over promptly and recommend immediate consumption; do not build an ambient ready-wrapped stock.

Troubleshooting: Soft crust: steaming sealed box; uneven cook: mixed piece sizes; low contribution: unweighed chicken.

Substitution: Bone-in chicken needs a separate raw-weight, yield and cook validation; this cost is for boneless portions.

Release check: weigh ingredients and 5 portions; log actual usable yield and rejects; compare centre, colour and texture to the target; confirm required temperatures and allergen label; record packaging and unsold stock.

| Batch ingredient cost | Saleable units | Ingredients/unit | Pack/unit | Other variable/unit | Test price | Contribution/unit |
|---|---|---|---|---|---|---|
| 6,362 | 5 | 1,272.3 | 100 | 100 | 2500 | 1,027.7 |

All values in FCFA. Contribution margin 41.1%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 773.2 FCFA. At 80% sell-through it falls to 659.6 FCFA per sold unit under the section 7 model.

# P18 Chicken vegetable samosas

Quality target: Thin crisp triangular shell, generous dry filling and fully sealed corners.

Pilot: 50 saleable units. Planning production batch: 250 units. Portion: 50 x 30 g wrapper + 40 g filling + 2 g paste; about 65–72 g cooked. Scale all components x5; validate capacity and losses before adoption.

| Prepared component or ingredient | Pilot g | Production g |
|---|---|---|
| B29 Samosa wrapper dough | 1,510 | 7,550 |

| Prepared component or ingredient | Pilot g | Production g |
|---|---|---|
| B31 Chicken vegetable samosa filling | 2,000 | 10,000 |
| B30 Samosa sealing paste | 100 | 500 |
| Oil | 200 | 1,000 |

1. Make B29 and two B31 batches. Weigh fifty 30 g wrappers, 40 g chilled filling and 2 g sealing paste each; wrapper spare 10 g.
2. Fold into cones, fill without air pockets and seal firmly. B30 must be scaled to supply 100 g usable paste, slightly above one base batch.
3. Fry at a trial 170–175°C for 4–6 minutes until crisp and filling meets S1 reheating control.
4. Drain and use H holding. For catering, agree a timed handover and holding method rather than leaving trays at ambient temperature.

Holding and packaging: H — The final bake/fry reheats the previously cooked chilled filling or topping. Serve promptly or hold hot under S1, with a house quality target of at most 2 hours. Do not cool finished leftovers for another reheat. Follow the product-specific transport and finishing route in section 5A.

Troubleshooting: Blisters/leaks: poor seal or wet filling; tough wrapper: too thick; greasy shell: low oil temperature.

Substitution: Vegetable version can replace chicken with cooked lentils only as a separately costed and labelled new filling.

Release check: weigh ingredients and 50 portions; log actual usable yield and rejects; compare centre, colour and texture to the target; confirm required temperatures and allergen label; record packaging and unsold stock.

| Batch ingredient cost | Saleable units | Ingredients/unit | Pack/unit | Other variable/unit | Test price | Contribution/unit |
|---|---|---|---|---|---|---|
| 5,315 | 50 | 106.3 | 10 | 25 | 300 | 158.7 |

All values in FCFA. Contribution margin 52.9%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 137.4 FCFA. At 80% sell-through it falls to 123.4 FCFA per sold unit under the section 7 model.

# P19 Loaded chicken shawarma fries

Quality target: Hot crisp fries, evenly distributed chicken and restrained sauces.

Pilot: 5 saleable units. Planning production batch: 25 units. Portion: 180 g fries + 80 g chicken + 25 g garlic sauce + 5 g hot sauce + 15 g pickle. Scale all components x5; validate capacity and losses before adoption.

| Prepared component or ingredient | Pilot g | Production g |
|---|---|---|
| B24 Crisp fries | 900 | 4,500 |
| B18 Cooked chicken shawarma meat | 400 | 2,000 |
| B20 Garlic mayonnaise sauce | 125 | 625 |

| Prepared component or ingredient | Pilot g | Production g |
|---|---|---|
| B21 Cooked hot sauce | 25 | 125 |
| B23 Refrigerator cucumber pickles | 75 | 375 |

1. Finish five portions of 180 g fries close to service.
2. Add 80 g freshly cooked or safely held B18 chicken to each, then 25 g B20, 5 g B21 and 15 g drained B23.
3. Use a wide shallow tray; sauce immediately before handover.
4. Never use this product to dispose of meat outside its safe time/temperature controls.

Holding and packaging: I — Assemble to order: keep hot and cold components separately controlled under S1 until assembly. Hand over promptly and recommend immediate consumption; do not build an ambient ready-wrapped stock.

Troubleshooting: Soggy base: over-saucing or deep closed box; uneven value: unweighed chicken; lukewarm fries: early assembly.

Substitution: Offer mild with sauce on the side; removing hot sauce alone does not materially reduce cost.

Release check: weigh ingredients and 5 portions; log actual usable yield and rejects; compare centre, colour and texture to the target; confirm required temperatures and allergen label; record packaging and unsold stock.

| Batch ingredient cost | Saleable units | Ingredients/unit | Pack/unit | Other variable/unit | Test price | Contribution/unit |
|---|---|---|---|---|---|---|
| 4,480 | 5 | 895.9 | 90 | 60 | 1800 | 754.1 |

All values in FCFA. Contribution margin 41.9%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 574.9 FCFA. At 80% sell-through it falls to 492.6 FCFA per sold unit under the section 7 model.

# P20 Condensed milk soaked cake portions

Quality target: Moist evenly soaked sponge that holds its shape, with no large pool of liquid.

Pilot: 12 saleable units. Planning production batch: 60 units. Portion: 950 g baked sponge + 800 g soak; 12 portions about 140 g after trim. Scale all components x5; validate capacity and losses before adoption.

| Prepared component or ingredient | Pilot g | Production g |
|---|---|---|
| B32 Milk cake sponge batter | 950 | 4,750 |
| B33 Three milk soak | 800 | 4,000 |

1. Bake a full B32 sponge and cool. Make or allocate 800 g B33; a full soak base leaves 190 g.
2. Prick sponge evenly, apply the weighed cold soak gradually, cover and refrigerate immediately.
3. Chill 4 hours as a quality trial; trim and divide into twelve portions. Gross cake mass 1,750 g less 70 g planned trim leaves 1,680 g, or 140 g each.

4. Sell from a refrigerator under holding class C. No whipped topping is included; add one only with a full formula, cost and cold-chain plan.

Holding and packaging: C — Chilled dairy cake: display and transport at or below 5°C. House pilot policy is prepare for same-day sale and discard unsold portions at close. This is a conservative operating policy, not a validated shelf life.

Troubleshooting: Wet puddles: too much soak or weak sponge; dry centre: uneven perforation; collapse: underbaked sponge.

Substitution: Use UHT milk and cream as specified; evaporated milk substitutions need solids/sweetness recalculation.

Release check: weigh ingredients and 12 portions; log actual usable yield and rejects; compare centre, colour and texture to the target; confirm required temperatures and allergen label; record packaging and unsold stock.

| Batch ingredient cost | Saleable units | Ingredients/unit | Pack/unit | Other variable/unit | Test price | Contribution/unit |
|---|---|---|---|---|---|---|
| 4,010 | 12 | 334.2 | 45 | 35 | 650 | 235.8 |

All values in FCFA. Contribution margin 36.3%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 169.0 FCFA. At 80% sell-through it falls to 132.3 FCFA per sold unit under the section 7 model.

# 5 Complete shawarma production and service

For the requested oven-cook, chill, portion and reheat route, use section 5A together with B18/B19 below. The vertical-spit workflow later in this section is a separate alternative, not a required step after oven cooking.

P13 and P14 are complete standard-size recipes. The smaller entry wrap below is a separate portion test, not a reduction hidden inside the standard product. A second shawarma machine is not confirmed: cook only one cone species on the existing machine at a time, or use a separate appropriate pan/griddle with utensils and raw/cooked separation. Do not stack chicken and beef together.

## B17 Shawarma flatbread

Pilot input 1,681 g; target usable output 1,500 g. Planning production batch is x5: input 8,405 g and usable output 7,500 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Bread flour | 1,000 | 5,000 | 100.00% |
| Water | 610 | 3,050 | 61.00% |
| Sugar | 15 | 75 | 1.50% |
| Salt | 18 | 90 | 1.80% |
| Yeast | 8 | 40 | 0.80% |
| Oil | 30 | 150 | 3.00% |

Baker's percentages use 1,000 g total flour as 100%; added milk/egg/fat are weighed ingredients, not equivalent to water.

1. Mix all ingredients 3 minutes slowly then 5–7 minutes until smooth and extensible.
2. Rest covered 30 minutes, divide into sixteen pieces of about 104 g, then rest another 15 minutes.
3. Roll each evenly to about 24–26 cm. Cook on a very hot clean plate or in a 240–260°C trial oven until set and just blistered, roughly 1–3 minutes; avoid crisping.
4. Stack briefly under a clean cover to retain flexibility, then cool without condensation. Target 1,500 g cooked bread; sixteen wraps at 90 g consume 1,440 g.

Yield and process note: Input 1,681 g; allow about 17 g handling and 164 g baking loss. Extra 60 g cooked bread is allocated to QC/weight variation in product costing.

Illustrative ingredient cost: 768 FCFA per pilot batch; 512.2 FCFA/kg usable output. This excludes packaging and operating costs.

## B18 Cooked chicken shawarma meat

Preparation-ahead option: section 5A gives the full oven, cooling, portioning and reheating method. Its final-yield adjustments apply when this base is cooked, chilled and reheated.

Pilot input 1,304 g; target usable output 750 g. Planning production batch is x5: input 6,520 g and usable output 3,750 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Chicken boneless | 1,000 | 5,000 | — |
| Yogurt plain | 150 | 750 | — |
| Lemon juice | 35 | 175 | — |
| Vinegar 5% | 15 | 75 | — |
| Oil | 30 | 150 | — |
| Garlic peeled | 25 | 125 | — |
| Salt | 16 | 80 | — |
| Paprika | 12 | 60 | — |
| Cumin | 6 | 30 | — |
| Coriander | 6 | 30 | — |
| Black pepper | 3 | 15 | — |
| Cinnamon | 1 | 5 | — |
| Chili trimmed | 5 | 25 | — |

1. Use trimmed boneless thigh; keep raw meat at or below 5°C. Mix marinade separately, coat meat, cover, label and marinate refrigerated 6–12 hours as a trial quality window.

2. Drain only excess free marinade. Weigh the raw load. Use the vertical machine only once its minimum cone size and safe operating instructions are established; otherwise cook small batches on a suitable hot plate.
3. Follow the dedicated shawarma service workflow in section 5. Shave only the cooked outer layer, then finish slices on the griddle and probe to S1 poultry temperature.
4. Target 750 g saleable cooked meat per 1,000 g purchased boneless meat. All marinade cost is charged even though some is discarded. Never reuse raw marinade as sauce.

Yield and process note: Gross input 1,304 g including marinade; 750 g cooked saleable output is an assumed purchase-to-cooked yield of 75%, not 75% of the marinated total.

Illustrative ingredient cost: 4,743 FCFA per pilot batch; 6,323.9 FCFA/kg usable output. This excludes packaging and operating costs.

## B19 Cooked beef shawarma meat

Preparation-ahead option: section 5A gives the full oven, cooling, portioning and reheating method. Its final-yield adjustments apply when this base is cooked, chilled and reheated.

Pilot input 1,231 g; target usable output 700 g. Planning production batch is x5: input 6,155 g and usable output 3,500 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Beef boneless | 1,000 | 5,000 | — |
| Onion trimmed | 80 | 400 | — |
| Lemon juice | 30 | 150 | — |
| Vinegar 5% | 30 | 150 | — |
| Oil | 30 | 150 | — |
| Garlic peeled | 20 | 100 | — |
| Salt | 16 | 80 | — |
| Paprika | 10 | 50 | — |
| Cumin | 6 | 30 | — |
| Coriander | 5 | 25 | — |
| Black pepper | 3 | 15 | — |
| Cinnamon | 1 | 5 | — |

1. Thinly slice trimmed beef across the grain. Combine marinade and coat in a covered food-safe container.
2. Marinate refrigerated 6–12 hours as a pilot quality window. Do not add more acid to tenderize an unsuitable tough cut indefinitely.
3. Cook using the same staged shawarma workflow as chicken; finish shaved slices thoroughly under S1's shawarma house standard.

4. Target 700 g cooked saleable beef per kilogram purchased boneless meat; record actual yield and discard raw marinade.

Yield and process note: Gross input 1,231 g; cooked saleable yield assumes 70% of raw purchased beef. It is a costing assumption, not a guaranteed yield.

Illustrative ingredient cost: 4,962 FCFA per pilot batch; 7,089.1 FCFA/kg usable output. This excludes packaging and operating costs.

# B20 Garlic mayonnaise sauce

Pilot input 987 g; target usable output 975 g. Planning production batch is x5: input 4,935 g and usable output 4,875 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Mayonnaise pasteurized | 800 | 4,000 | — |
| Lemon juice | 65 | 325 | — |
| Water | 65 | 325 | — |
| Garlic peeled | 50 | 250 | — |
| Salt | 5 | 25 | — |
| Black pepper | 2 | 10 | — |

1. Use commercial pasteurized mayonnaise; do not prepare a raw-egg mayonnaise.
2. Blend garlic with lemon and water using sanitized tools, then whisk into mayonnaise and season.
3. Weigh and fill small covered service bottles; label and refrigerate under S1.
4. Use a clean bottle for each replenishment rather than topping fresh sauce into old sauce.

Yield and process note: Input 987 g; 12 g bowl loss. Water and garlic addition invalidate the original unopened mayonnaise shelf life.

Illustrative ingredient cost: 2,287 FCFA per pilot batch; 2,346.0 FCFA/kg usable output. This excludes packaging and operating costs.

# B21 Cooked hot sauce

Pilot input 880 g; target usable output 700 g. Planning production batch is x5: input 4,400 g and usable output 3,500 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Tomato trimmed | 500 | 2,500 | — |
| Chili trimmed | 120 | 600 | — |
| Onion trimmed | 100 | 500 | — |
| Garlic peeled | 20 | 100 | — |

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Vinegar 5% | 80 | 400 | — |
| Oil | 30 | 150 | — |
| Sugar | 20 | 100 | — |
| Salt | 10 | 50 | — |

1. Soften onion and garlic in oil, add chopped tomato, chili, vinegar, sugar and salt.
2. Simmer 15–20 minutes, blend safely and return to a bubbling simmer; reduce to 700 g.
3. Cool actively under S1, bottle hygienically and refrigerate.
4. Offer a measured 5 g serving separately to let customers choose heat.

Yield and process note: Input 880 g; 180 g evaporation. This is not a shelf-stable bottled sauce; acidity is unvalidated.

Illustrative ingredient cost: 920 FCFA per pilot batch; 1,314.3 FCFA/kg usable output. This excludes packaging and operating costs.

## B22 Shawarma slaw

Pilot input 995 g; target usable output 975 g. Planning production batch is x5: input 4,975 g and usable output 4,875 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Cabbage trimmed | 700 | 3,500 | — |
| Carrot trimmed | 180 | 900 | — |
| Onion trimmed | 60 | 300 | — |
| Vinegar 5% | 35 | 175 | — |
| Sugar | 15 | 75 | — |
| Salt | 5 | 25 | — |

1. Wash trimmed vegetables with potable water and use sanitized ready-to-eat preparation tools.
2. Shred finely, mix dressing, toss and drain excess free liquid briefly.
3. Weigh finished yield and chill covered. Portion only small quantities onto the service line.
4. Serve 45 g per wrap. Keep separate from raw poultry, its containers and its wash area.

Yield and process note: Input 995 g; 20 g draining/handling loss. Vinegar does not replace refrigeration.

Illustrative ingredient cost: 676 FCFA per pilot batch; 692.8 FCFA/kg usable output. This excludes packaging and operating costs.

# B23 Refrigerator cucumber pickles

Pilot input 1,480 g; target usable output 650 g. Planning production batch is x5: input 7,400 g and usable output 3,250 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Cucumber trimmed | 700 | 3,500 | — |
| Vinegar 5% | 350 | 1,750 | — |
| Water | 350 | 1,750 | — |
| Sugar | 60 | 300 | — |
| Salt | 20 | 100 | — |

1. Dissolve sugar and salt in the water with heat, add vinegar, and cool the brine rapidly before adding clean sliced cucumber.
2. Submerge in a clean covered food-safe container and refrigerate overnight for flavour.
3. Drain only the service quantity; target 650 g drained cucumber from the batch. Brine is discarded and fully charged to cost.
4. Label as refrigerated quick pickles. No ambient-storage or canning claim is made.

Yield and process note: Gross input 1,480 g including brine. Only drained pickles are costed as usable output.

Illustrative ingredient cost: 933 FCFA per pilot batch; 1,435.0 FCFA/kg usable output. This excludes packaging and operating costs.

# B24 Crisp fries

Pilot input 1,128 g; target usable output 650 g. Planning production batch is x5: input 5,640 g and usable output 3,250 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Potato peeled | 1,000 | 5,000 | — |
| Oil | 120 | 600 | — |
| Salt | 8 | 40 | — |

1. Cut peeled potatoes to uniform 8–10 mm sticks. Rinse with potable water, drain and dry thoroughly away from hot oil.
2. For immediate service, trial a first fry around 150–160°C until cooked but pale, then finish around 175°C until crisp. Keep batches small.
3. Drain on a rack and salt. Serve promptly; do not hold a large tray at room temperature for later frying.
4. If par-frying in advance, use the controlled cooling and refrigerated storage workflow S1, then finish to order.

Yield and process note: 650 g assumed saleable fries per kg peeled potato. Oil entry is a 120 g usage/disposal cost allowance, not a claim that all oil is absorbed. Cost the fryer charge separately as working stock.

Illustrative ingredient cost: 1,095 FCFA per pilot batch; 1,684.9 FCFA/kg usable output. This excludes packaging and operating costs.

## B35 Beef shawarma sesame sauce

Pilot input 990 g; target usable output 975 g. Planning production batch is x5: input 4,950 g and usable output 4,875 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Mayonnaise pasteurized | 500 | 2,500 | — |
| Tahini | 200 | 1,000 | — |
| Lemon juice | 80 | 400 | — |
| Water | 180 | 900 | — |
| Garlic peeled | 25 | 125 | — |
| Salt | 5 | 25 | — |

1. Whisk tahini with lemon and water until smooth, then incorporate mayonnaise, garlic and salt.
2. Use sanitized utensils and check supplier sesame and egg allergen declarations.
3. Refrigerate in small labelled service bottles under S1.
4. Portion 30 g per beef wrap; keep a separate spoon for sesame sauce.

Yield and process note: Input 990 g; 15 g handling loss. A garlic-only alternative uses B20 at the same serving mass; re-cost.

Illustrative ingredient cost: 2,872 FCFA per pilot batch; 2,945.5 FCFA/kg usable output. This excludes packaging and operating costs.

## Standard and small wrap assembly

| Component | Standard chicken g | Standard beef g | Small chicken g | Small beef g |
|---|---|---|---|---|
| Cooked flexible bread | 90 | 90 | 70 | 70 |
| Cooked meat | 110 | 110 | 80 | 80 |
| Cooked fries | 50 | 50 | 35 | 35 |
| Slaw | 45 | 45 | 30 | 30 |
| Drained pickles | 15 | 15 | 10 | 10 |
| Garlic or sesame sauce | 30 | 30 | 20 | 20 |
| Optional hot sauce, included in cost | 5 | 5 | 3 | 3 |

| Component | Standard chicken g | Standard beef g | Small chicken g | Small beef g |
|---|---|---|---|---|
| Approximate finished weight | 345 | 345 | 248 | 248 |

Small bread: portion B17 dough into approximately 80 g pieces and cook to about 70 g flexible wraps; validate diameter and measured bake loss. Use the same component recipes for both sizes. A standard chicken portion requires 110/0.75 = 146.7 g raw boneless purchase weight; standard beef requires 110/0.70 = 157.1 g. The smaller portion needs 106.7 g raw chicken or 114.3 g raw beef. These assumptions include no bone yield.

Small chicken wrap example: ingredient cost 686.7 FCFA; packaging 65 and other variable allowance 45 FCFA; price 1,500 FCFA; contribution 703.3 FCFA before fixed costs. This assumes exactly usable component weights and must add measured bread/assembly rejects in practice.

Small beef wrap example: ingredient cost 759.9 FCFA; packaging 65 and other variable allowance 45 FCFA; price 1,800 FCFA; contribution 930.1 FCFA before fixed costs. This assumes exactly usable component weights and must add measured bread/assembly rejects in practice.

# Vertical spit service workflow

1. Before service: verify raw storage, refrigerated garnish station, handwashing, cleaned probes, clean knives, ventilation and machine guards. Make a cone only as large as forecast sales and the manufacturer’s minimum allow; switch to small-batch pan cooking if those quantities conflict.
2. Keep raw marinated meat cold until loading. Cook continuously according to the machine instructions; do not turn off heat while leaving a partly cooked cone warm for later service. Never add raw meat on top of a cooking cone.
3. Shave thin slices only from a thoroughly cooked exterior using the designated knife. Finish sliced meat immediately on the hot plate; verify the S1 house centre-temperature target. Exterior colour alone is insufficient.
4. Transfer cooked meat with clean utensils into a controlled hot holding pan or directly into a wrap. Never use the raw-meat tray or tongs for finished slices.
5. At close, use the conservative one-day cone policy: discard unsold cone and shaved meat. Do not refrigerate a partly cooked cone for the next day. Reduce tomorrow’s starting cone size if waste is excessive. Local-authority doner safety guidance
6. Clean and sanitize raw station, knife, drip areas, hot plate and service surfaces according to equipment and chemical instructions; record disposal and close temperatures.

# Station and handover

Flow: refrigerated raw meat → dedicated cooker → clean finishing/hot-holding zone → wrap assembly → toast → packaging → customer. Beside assembly, keep slaw, pickles and sauces refrigerated with separate utensils. Put raw-meat prep physically away from ready-to-eat vegetables. One person handles money or washes hands and changes tasks before touching food. Use measured scoops verified on scales, not free pouring.

Service target after training: a standard wrap in 3–5 minutes when safe components are ready. During low demand, cook to order in smaller batches rather than maintaining an oversized cone. Optional 20 g cheese

adds 130 FCFA ingredient cost at the assumed price before other costs; trial an add-on price of 250–300 FCFA. Do not add optional components without a portion and allergen instruction.

# 5A Prepare ahead: burgers, meat, chicken and fish

## Decision for this bakery

Yes: fully cook suitable meat at the production site, cool it rapidly, weigh and label chilled portions, transport them cold, and reheat only the portions ordered before assembly. This can suit the two-site business and avoids keeping a large shawarma cone running during quiet periods. It is a complete cooking and controlled cooling process; briefly warming cooked meat or toasting an assembled cold burger is insufficient.

This revision retains the original 20 core products and eight frozen formats, and adds four optional savoury products A01–A04. They are development options, not a recommendation to launch 32 products together. S1 and this section govern preparation ahead wherever an older recipe refers to cooking, cooling, transport or reheating. All cooking times, tray loads and yields below require a small measured trial on the actual oven.

## Product-by-product service decision

| Product | What to prepare ahead | What happens when ordered or in a small service batch |
|---|---|---|
| P13/P14 chicken/beef shawarma | Fully oven-cooked strips, cooled and portioned; bread, sauces and garnish stored separately | Reheat meat thoroughly, cook fries, assemble and toast bread briefly |
| P16 beef cheeseburger | Fully cooked individual patties, rapidly cooled separately | Reheat patty, melt cheese, toast bun, then add cold garnish and sauce |
| P15 crispy chicken burger; P17 chicken box | Chilled raw portions and separate coating; or fully cooked chilled chicken after a texture trial | Prefer a full fry to order; the alternative is one complete reheat in a dry oven or suitable fryer, with temperature and crispness checks |
| P06 fish roll; P07 beef pie; P09 chicken pie; P18 chicken samosa | Cooked chilled filling and separate dough; assemble small chilled batches | Final fry/bake heats the filling again: this is its one reheating step. Serve or hold hot; do not chill finished leftovers for another reheat |
| P08 sausage roll | Chilled raw sausage filling enclosed in pastry | Bake completely as the first cook; a finished roll can instead be cooled and reheated once under the documented route |
| P10 egg roll | Cooked chilled egg and separate dough | Final fry is the egg's reheating step; serve or hold hot, with no later cool-and-reheat cycle |
| P12 chicken pizza; A04 beef pizza | Chilled cooked topping, dough and sauce kept separate | Final pizza bake is the meat's reheating step; verify the topping temperature and serve |
| P19 loaded chicken fries | Chilled cooked chicken; sauces and pickles separate | Reheat chicken once, cook fries fresh, assemble immediately |
| A01 roast chicken burger | Oven-cooked chicken portion | Reheat, then assemble with toasted bun and chilled garnish |

| Product | What to prepare ahead | What happens when ordered or in a small service batch |
|---|---|---|
| A02 fish burger; A03 fish wrap | Deboned raw chilled fish for cooking to order, or fully cooked chilled fish for the measured oven route | Cook or reheat as applicable, then assemble; avoid repeated handling of fragile fish |

The single-reheat house rule follows UK food business guidance. It is an operating choice for this plan, not a statement of Cameroon law. Track the heat history of the ingredient: putting reheated chicken into a different product does not reset its history. Prefer fully cooked patties and roast chicken for the initial prepare-ahead trial; compare fish and crispy products with freshly cooked controls before adoption.

# One controlled route from production to the customer

1. Receive sound ingredients and keep raw proteins chilled under S1. Thaw covered in the refrigerator, below ready-to-eat food. Weigh raw portions with dedicated utensils; wash hands and clean/sanitize the area before handling cooked food. Marinade is not a preservation treatment. Discard raw-contact marinade after use; never use it as a finishing sauce.

2. Fully cook using the appropriate method below. Use the S1 house cooking target throughout; check the thickest portions at the oven's slowest-heating positions. Record batch identity, time and measured food temperature. Browning is a quality cue, not evidence of core doneness. Never partly cook meat or chicken, refrigerate it, then finish later. USDA guidance on partial cooking

3. Start active cooling immediately after cooking. Arrange patties or fillets in a single layer; spread strips/fillings in shallow food-safe pans with circulation around them. Use a blast chiller if available, or another method demonstrated to meet S1. A large refrigerator/freezer does not establish sufficient cooling capacity. Protect loosely covered food during cooling, then cover fully when chilled. Log the cooling checkpoints; do not leave trays on the counter to reach room temperature. FDA employee cooling guidance

4. Weigh chilled usable output with clean utensils, quickly enough to maintain temperature control. Portion into shallow labelled containers; do not pack hot food into stacked deep takeaway boxes. Label recipe, species, batch, cook time, cooling completion, weight, use-by and reheat count. Patties may be formed before cooking; weigh after chilling to check shrinkage. Discard bones, gristle and unsuitable pieces before calculating saleable yield.

5. Transport covered portions with sufficient ice packs in clean insulated carriers, or suitable refrigerated transport. Record dispatch and arrival food temperatures; maintain the S1 cold limit during the journey and sales-site storage. Keep raw and cooked foods separate. Do not rely on journey length alone. Quarantine and discard food with an unknown unsafe temperature history; reheating cannot repair every form of temperature abuse.

6. Reheat only the ordered portion, directly from refrigeration, using a suitable induction pan with a lid, griddle or small oven. Reheat rapidly to the S1 house target, confirmed at the thickest centre; a holding cabinet is not a reheater. Avoid using the large rotary oven for individual orders. Set a timed method only after testing multiple portions and starting temperatures. Add any measured drinking water early enough for the complete portion to reach the target; do not add cold juices afterwards.

7. Toast the bun or bread separately. Add the hot protein, then refrigerated sauce, washed garnish and drained pickles with clean utensils. Serve promptly. A brief final wrap toast is a bread-quality step after

the meat is already safely reheated. Never prepare a stock of warm assembled burgers or wraps on the counter.

8. For a small rush, maintain finished hot protein under S1; monitor actual food temperature. Use the report's two-hour hot-display quality target and reduce batches if drying occurs. Reheated unsold portions are discarded, not cooled again. Initial launch policy remains same-day preparation and sale with disposal at close. Do not infer a refrigerated multi-day shelf life from this process. Planned frozen production needs its own documented storage, thawing and quality trial before rollout; freezing must never be used to rescue mishandled food.

# Oven method for B18 chicken and B19 beef shawarma

Use the complete marinades already specified in B18/B19. For a pilot use their 1 kg raw boneless meat formulas; multiply every ingredient by five only after proving the smaller load. Bone-in purchasing requires an additional measured deboning yield. Food-grade solid roasting pans with raised edges are needed inside the 40 × 60 cm trays; do not let raw juices drip through racks. Do not bake bread or ready-to-eat pastries underneath or alongside raw proteins.

Chicken: use boneless thigh for a forgiving reheating trial, or test breast separately because it dries more easily. Cut even pieces about 15–20 mm thick, marinate covered under refrigeration for a trial 2–12 hours within the same-day workflow, and spread in one layer without crowding. Trial a preheated 190°C oven, checking early and typically around 15–25 minutes depending on cut and loading. Turn or redistribute only with raw/cooked utensil control. Release by the S1 probe result, not time. A lightly browned surface with moist interior is the quality target. Slice using a clean knife after cooking, then cool promptly; include slicing time in the cooling record.

Beef: choose a tender boneless cut and slice across the grain into even thin strips. Use B19 marinade and a short refrigerated trial of 2–6 hours. Trial 190–200°C in a shallow single layer, starting checks around 8–15 minutes. The cook may be shorter or longer; do not hold it in the oven merely to reach an assumed time. Tough stew cuts are not interchangeable with tender strips: use a separately tested covered braise, then cool and portion the shredded result with a revised yield. Do not promise shawarma texture from an unidentified cheap cut.

The original B18/B19 targets of 750/700 g per 1 kg raw meat are initial cooking assumptions. For the new route, weigh output after cooling and again after reheating. Use only the final usable meat mass for the advertised serving. If the chilled yield matches those assumptions and reheating retains 95%, final yields become 712.5 g chicken and 665 g beef. Do not add raw marinade weight to saleable meat yield. Reheat with only a small measured splash of water when needed; avoid boiling strips in a large pool of liquid.

# Beef hamburger patties: oven, portion and finish

Keep B27's full formula: beef mince 1,000 g; salt 12 g; black pepper 3 g; paprika 3 g; oil allowance 10 g. Mix the mince and dry seasoning briefly; use the oil for pan release. The meat mixture is 1,018 g before the pan oil. Form ten equal patties approximately 102 g each, about 12–15 mm thick. Do not heavily compress or press them during cooking. Keep raw patties cold until loading.

Trial cook on lightly oiled raised-edge pans at 190°C, turning once if needed, with initial checks around 10–15 minutes. Probe sideways into the centre, reach S1, then cool patties separately in one layer. This fully cooked oven route is an alternative to B27's fresh hot-plate method. At service, reheat the patty in a suitable pan/oven,

check the centre, then melt the 20 g cheese portion and assemble P16. A quick surface sear alone does not adequately reheat the centre.

P16 promises 75 g final cooked beef. Its original 750 g batch assumes ten portions directly after cooking. At an illustrative 95% reheating retention, a 75 g chilled patty yields only 71.25 g after reheating. To preserve the 75 g final claim, target 78.95 g chilled per patty and scale the B27 batch by 1/0.95: beef 1,052.6 g; salt 12.63 g; pepper 3.16 g; paprika 3.16 g; oil 10.53 g. Form ten larger raw patties from that mixture and validate the final yield. Do not silently describe a smaller reheated portion as 75 g. Whole patties cannot be topped up neatly with scraps; adjust raw forming weight from measured losses.

An alternative is to chill safely prepared raw patties and cook fully to order. Compare that route with prepare-ahead patties for juiciness, waiting time, saleable yield and total cost. Never pre-brown raw patties for later completion.

## Crispy chicken, pies, rolls and samosas

For B26/P15/P17, the best starting quality trial is refrigerated raw marinated portions followed by coating and a full fry to order. Keep coating contaminated by raw chicken away from clean dry stock and discard the used portion. Do not partly fry chicken and store it for later finishing. A fully cooked, rapidly chilled portion may be reheated once in a dry oven or suitable fryer, but the centre must meet S1; reject a method that burns the outside, dries the meat or softens the coating. Do not assume the existing 1,150 g coated yield remains unchanged after this extra process.

For P06/P07/P09/P18, cool the prepared filling before assembly. The final bake/fry reheats that filling. If production and sales are separate, either transport the finished product hot under control, or transport properly chilled assembled uncooked pastries for final cooking in suitable equipment at the sales site. The latter route needs validated same-day handling and final cooking capacity; no sales-site oven has been confirmed. If neither route is possible, postpone that product rather than chilling a finished twice-heated filling for a third heat cycle. P10 cooked egg follows the same principle.

P08 is different because B10 uses raw sausage meat. Baking the assembled sausage roll is its first complete cook; it may then be rapidly cooled and reheated once before sale, with both stages checked under S1. Do not substitute an already cooked sausage without revising that history. Reheated pastry should be heated in an oven appropriate to the sales site for texture; a warm display unit cannot perform this step.

## Additional complete recipe B36: lemon-garlic oven fish

This is a new shared base for A02/A03. Use supplier-identified, food-grade, boneless tilapia or another firm white-fish fillet. The recipe is for trimmed edible fillet, not whole fish. Inspect carefully for pin bones before and after cooking. Keep fish utensils separate; fish is an additional allergen in the burger/wrap station.

| Ingredient | Pilot g | Production trial g (×5) |
|---|---|---|
| Boneless trimmed white fish | 1,000 | 5,000 |
| Oil | 25 | 125 |
| Lemon juice | 25 | 125 |
| Garlic, finely crushed | 15 | 75 |
| Salt | 10 | 50 |

| Ingredient | Pilot g | Production trial g (×5) |
|---|---|---|
| Paprika | 4 | 20 |
| Black pepper | 2 | 10 |
| Total input | 1,081 | 5,405 |

1. Keep fillets chilled, drain excess thaw water and pat with disposable food-safe towel. Cut pieces of similar thickness, roughly 15–20 mm; keep smaller pieces in a separate trial group. Mix the listed seasonings and oil, coat fish, and refrigerate for a short trial of 15–20 minutes. Acid does not make raw fish safe; extended lemon contact can soften it.
2. Place in a single layer in a lined raised-edge pan. Trial roast at 180–190°C, checking around 10–16 minutes; thickness and oven loading determine the actual duration. Meet S1 at the thickest centre. Target moist flakes and light browning, with no raw centre. Inspect for remaining bones with clean utensils without breaking every piece unnecessarily.
3. For immediate service, portion after cooking and assemble. For preparation ahead, cool, portion and transport using the controlled route above. Trial reheating covered briefly to limit drying, then uncover if surface moisture needs removal; meet S1 before assembly. Do not mix yesterday's fish with a new batch or use strongly deteriorated fish with extra spice.
4. Planning yield: 800 g chilled usable fish per pilot, then 760 g after a trial 95% reheating retention. These are unverified assumptions, not measured performance. Record cooking, chilling, trimming and reheating losses separately. At 120 g final fish, one pilot supports six sale portions and a 40 g final-equivalent balance; at 100 g, seven and a 60 g balance. Balances must have an identified same-day use or be charged as waste, not counted as full portions.

Illustrative raw fillet price: 4,500 FCFA/kg edible, newly assumed for costing only. Other inputs use section 7 prices. B36 pilot ingredients cost **4,684.0 FCFA**, or **6,163.2 FCFA/kg of final reheated output**. Whole-fish purchase cost per usable kg equals whole-fish price divided by measured fillet yield, before cooking losses. Supplier pricing and bone removal can materially change viability.

# A01 Roast chicken burger — ten-unit pilot

Use B25 buns and B28 sauce exactly as formulated earlier. This is an oven-roasted alternative to P15's crispy chicken.

| Component | Final amount per burger g | Ten-unit pilot g |
|---|---|---|
| B25 baked bun | 87 | 870 |
| B18 chicken, after final reheat | 110 | 1,100 |
| B28 burger sauce | 25 | 250 |
| Lettuce | 15 | 150 |
| Tomato | 25 | 250 |
| B23 drained pickles | 10 | 100 |
| Finished assembly | 272 | 2,720 |

At 95% reheating retention, prepare 1,157.9 g chilled B18 chicken: multiply every ingredient in the B18 750 g base by 1.54386 (raw boneless chicken 1,543.9 g). Weigh ten chilled portions about 115.8 g, then confirm 110 g after reheating. Reheat chicken under S1, toast bun, spread sauce, add lettuce/tomato/pickles and assemble promptly. Choose a bun diameter that holds the chicken without excessive stacking. Keep sauce and vegetables refrigerated until assembly. For 50 burgers multiply every table quantity and base scaling factor by five after the pilot passes. Allergens: wheat/gluten, milk, egg and mustard in the specified bases; check purchased sauce labels.

## A02 Lemon-garlic fish burger — ten-unit pilot

| Component | Final amount per burger g | Ten-unit pilot g |
|---|---|---|
| B25 baked bun | 87 | 870 |
| B36 fish, after final reheat | 120 | 1,200 |
| B28 burger sauce | 25 | 250 |
| Lettuce | 15 | 150 |
| Tomato | 20 | 200 |
| B23 drained pickles | 10 | 100 |
| Finished assembly | 277 | 2,770 |

Multiply every B36 ingredient by 1,200/760 = 1.57895 (raw boneless fish 1,578.9 g). Chilled fish needed is 1,263.2 g, or about 126.3 g per portion, to target 120 g after reheating. Reheat gently but completely under S1; use a broad clean spatula to keep pieces intact. Toast bun, spread sauce, add fish and cold garnish, and hand over promptly. For cooking to order, use measured first-cook yield instead of adding an unnecessary reheat allowance. For 50 burgers multiply all pilot quantities by five after validation. Allergens: fish, wheat/gluten, milk, egg and mustard; segregate fish from the beef/chicken service tools.

## A03 Lemon fish wrap — ten-unit pilot

| Component | Final amount per wrap g | Ten-unit pilot g |
|---|---|---|
| B17 cooked flatbread | 90 | 900 |
| B36 fish, after final reheat | 100 | 1,000 |
| B22 cabbage carrot slaw | 45 | 450 |
| B23 drained pickles | 15 | 150 |
| B20 garlic sauce | 25 | 250 |
| B24 freshly cooked fries | 35 | 350 |
| B21 hot sauce, optional but costed | 5 | 50 |
| Finished assembly | 315 | 3,150 |

Multiply all B36 ingredients by 1,000/760 = 1.31579 (raw fillet 1,315.8 g); target 1,052.6 g chilled fish before reheating. Scale every other base by the pilot quantity divided by its listed usable yield. Warm the bread, reheat

fish under S1, cook fries fresh, then add weighed slaw, pickles and sauces. Fold gently so fish does not crush the bread; a short final toast is for the bread after the protein has met S1. Serve immediately. The ten portions consume exactly the listed prepared component amounts; any whole-base leftovers require planned same-day use or a waste charge. Scale to 50 wraps by multiplying every pilot quantity by five. Allergens: fish, wheat/gluten and egg/mustard as present in sauces; check milk in purchased ingredients.

## A04 Beef mini pizza — ten-unit pilot

| Component | Amount per pizza g | Ten-unit pilot g |
|---|---|---|
| B15 pizza dough | 150 | 1,500 |
| B16 pizza sauce | 40 | 400 |
| B19 beef, final target after pizza bake | 50 | 500 |
| Mozzarella | 50 | 500 |
| Onion | 10 | 100 |
| Sweet pepper | 10 | 100 |

Use 526.3 g chilled B19 beef at the illustrative 95% final-bake retention: scale every B19 ingredient by 526.3/700 = 0.75188 (raw beef 751.9 g). Do not separately reheat the beef and then store it again. Prepare ten 150 g dough rounds using B15, add 40 g sauce, 52.6 g cold cooked beef, 50 g cheese and 10 g each onion/pepper. Trial bake using P12's oven method and check that the meat topping reaches S1 during that same bake. The bake is the beef's one reheating step. Serve immediately or hold hot under S1; do not cool unsold pizzas for another reheat. Target approximately 275–290 g finished pizza, verified by weighing; moisture losses in dough and vegetables are separate from the meat allowance. Scale to 50 pizzas by multiplying all ingredients by five. Allergens: wheat/gluten and milk, plus anything declared in purchased sauces.

## Costing the preparation-ahead route

All new prices below are test assumptions in FCFA, not observed Nkambe prices or promised profit. Base cost includes all its ingredient inputs divided by usable yield. The additional 5% reheating loss is an illustrative planning allowance; replace it with measured chilled-to-final retention for each product. Prepared components in these ten-unit pilots are scaled proportionately; if staff make whole batches and discard balances, charge that extra waste separately.

| Product | Ingredients/unit | Pack | Other variable | Test price | Contribution | Margin |
|---|---|---|---|---|---|---|
| A01 Roast chicken burger | 917.7 | 60 | 100 | 2,200 | 1,122.3 | 51.0% |
| A02 Fish burger | 921.1 | 60 | 100 | 2,300 | 1,218.9 | 53.0% |
| A03 Fish wrap | 839.3 | 65 | 110 | 2,200 | 1,185.7 | 53.9% |
| A04 Beef mini pizza | 835.3 | 45 | 90 | 1,500 | 529.7 | 35.3% |

The variable allowance covers a provisional combined amount for cooking/reheating energy, cold handling, cleaning and incremental service work. It is not a measured quotation and excludes fixed payroll, rent, financing and taxes. Fixed overhead still has to be paid from contribution. Sales prices must be tested against portion size and local willingness to pay; the fish menu may be unsuitable at the assumed fillet price.

For the existing products, update rather than reuse the original margin unaltered. At unchanged final portions and 95% reheat retention, the protein ingredient surcharge alone is about **24.1 FCFA per P16 burger**, **36.6 per standard P13 chicken wrap**, **41.0 per standard P14 beef wrap**, and **26.6 per P19 loaded fries**. Add measured refrigeration, packaging, transport, energy and labour differences as well. These surcharges assume the original cooking yield equals the measured chilled yield; if it does not, recalculate from raw purchase weight. Pies/fillings and coated chicken need their own measured yields rather than a universal 5% deduction.

Cost per final kg of meat = total raw batch ingredient cost ÷ chilled usable kilograms ÷ reheating retention. Reheat retention = final edible weight ÷ chilled portion weight. Yield percentages multiply: chicken at 75% chilled yield and 95% reheat retention gives 71.25% final raw-to-service yield. Margin is not markup, and contribution is not net profit.

# Staff release sheet and equipment implications

| Record for every production batch | Entry needed |
|---|---|
| Identity and quantities | Date, recipe, supplier lot, raw kg, operator and intended products |
| Cooking | Oven/pan setting, tray positions, thickest-centre probe results and time |
| Cooling | Start and checkpoint times/temperatures against S1; pan depth and batch size |
| Portioning | Chilled usable kg, portion g, number packed, rejects and label/use-by |
| Transfer | Dispatch and receipt times/food temperatures, carrier and receiving operator |
| Service | Reheat result, final portion weight, reheat count, orders sold and disposal |
| Quality/cost | Juiciness or crispness, assembly time, measured retention, energy and total unit cost |

Begin with one or two trays, map cooking and cooling performance, and increase only when the slowest portions meet the controls. The 16-tray oven's physical capacity is not evidence that refrigeration can safely cool 16 trays at once. Separate production of different species where needed for handling, allergens and temperature checks. Use oven-safe pans, cleaned probe thermometers, accurate scales, shallow cooling containers, insulated transport carriers and a suitable sales-site reheating pan or oven. Confirm what is already owned before buying anything.

Cooking in the diesel oven uses diesel plus its listed 3 kW electrical auxiliaries. Refrigeration after cooking and reheating at the sales point still consume power. The sales inverter remains 15 kW with 30 kWh nominal battery storage; replacing the 3.5 kW shawarma appliance with a 5 kW induction cooker can increase instantaneous demand if operated at full power. Rebuild the simultaneous-load schedule with both ice-cream machines, fryers and refrigeration before service. Keep cold storage supplied continuously; this workflow does not justify switching it off to run another appliance.

Launch acceptance: a documented safe process, a successful cold-transport trial, repeatable final weights, acceptable taste against fresh controls, and a revised contribution calculation using actual quotations and losses. If cooling capacity or service reheating fails, reduce the batch or cook fully to order. Do not extend storage time to make an oversized batch commercially convenient.

# 6 Frozen desserts and two-size popsicles

The eight entries below are distinct sale formats or processes: soft serve, sundae, milkshake, sandwich, milk pop, fruit ice pop, yogurt bar and coated milk bar. Flavour variations are not counted as additional business concepts. There is no confirmed batch freezer, so this plan does not promise conventional churned hard ice cream production. Frozen sandwiches and coated bars use shallow hardening trials with the existing freezing equipment.

## F0 Frozen dairy processing and machine control

Gravity machines need a smooth mix that flows reliably. Use the actual manufacturer's approved range, hopper refrigeration requirement, minimum charge and cleaning instructions. The 1 kg formula is a composition pilot, not necessarily a sufficient operating charge. Separate hopper cooling and approved standby functions are model-specific; an unknown machine must not inherit another brand's operating procedure. Taylor gravity soft-serve example

For on-site dairy-mix processing, a published technical benchmark is to heat every part of the mix to at least 69°C for 30 minutes or 80°C for 25 seconds. This report uses the 80°C/25-second route only as a development reference for an agitated, measured and professionally validated process, not a claim that heating an ordinary pot once pasteurizes it. If even heating, hold-time control and rapid chilling cannot be demonstrated, use an approved ready-to-use pasteurized mix. University of Guelph dairy processing reference

After a validated heat step, chill promptly under S1 and target 1–4°C before aging/loading. Never put warm mix into the hopper. Keep pre-freeze mix covered and chilled, process it in small batches, and do not return dispensed ice cream to the hopper. Use the actual manual's dismantling, detergent, sanitizer concentration/contact time, rinsing and lubrication directions. The safe default launch routine is fresh mix each day, no carryover/rerun, and daily dismantle-clean-sanitize unless a documented approved cycle explicitly differs.

For frozen stock, adopt a storage target of −18°C or colder and log temperatures. This is an operating target, not a validated shelf life; a storage freezer's nameplate does not establish its ability to rapidly harden new bars. Pilot only small lots for sale within 7 days while checking flavour and texture; any longer commercial shelf life requires process and product validation. Soft serve is dispensed for immediate consumption, not stored as prefilled cups at its serving temperature.

## Measuring overrun and mould volume

Overrun measures added volume, not extra mass. Use: overrun % = 100 × (weight of a fixed cup of liquid mix / weight of the same cup of frozen product − 1). At assumed mix density 1.09 kg/L and 30% overrun, frozen density is 1.09/1.30 = 0.838 kg/L. Measure rather than assuming either machine reaches that value. University of Guelph overrun calculations

The owner confirms the large pop is twice the small one, but no absolute volume is known. Calibrate each mould with water to its intended fill line: tared water grams approximate ml. Repeat three times and average. Then weigh a known volume of actual mix for its density. Fill mass = working volume × mix density. Units = floor(usable mix mass / fill mass); remainder = usable mix mass − units × fill mass. Leave manufacturer-required expansion space; do not fill to the brim.

| Illustration only | Small | Large |
|---|---|---|
| Working fill volume | 50 ml | 100 ml |
| Mix density assumed | 1.10 g/ml | 1.10 g/ml |
| Fill mass | 55 g | 110 g |
| Usable mass from 1 kg formula | 970 g | 970 g |
| Full units | 17 | 8 |
| Unallocated remainder charged to batch | 35 g | 90 g |

For a x5 batch, 4,850 g usable mix would fill 88 small or 44 large example moulds, leaving 10 g; it does not have to preserve the inefficient pilot's 17/8 yields. The costing table deliberately uses conservative pilot counts. Recalculate production contribution using actual cavity count, startup residue and integer mould yields; do not mix the two bases when reporting profit.

## Frozen mix composition assumptions

For trial calculations, UHT whole milk is assumed 3.5% fat and 13% total solids; cream 35% fat and 40% solids; skim powder 97% solids; condensed milk uses section 4's assumed composition; glucose is dry glucose powder, not an 80% solids syrup. FBASE1 is approximately 6.2% fat and 35% total solids. These are engineering estimates to check against labels, not laboratory analyses or legal "ice cream" classifications.

Glucose form, fruit sugar, milk solids and stabilizer affect freezing and viscosity. Do not replace dry glucose with equal-weight syrup or omit stabilizer without rebalancing. FBASE3 uses pasteurized fruit ingredients and supplier-defined handling rather than borrowing a dairy pasteurization schedule. FBASE4's yogurt and pasteurized puree are assembled hygienically cold; the final mix has no claimed kill step. Keep both out of an unapproved dairy hopper.

## FBASE1 Gravity soft serve vanilla trial mix

Pilot input 1,000 g; target usable output 970 g. Planning production batch is x5: input 5,000 g and usable output 4,850 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| UHT milk | 660 | 3,300 | — |
| Cream 35% | 100 | 500 | — |
| Skim milk powder | 55 | 275 | — |
| Sugar | 100 | 500 | — |
| Glucose powder | 30 | 150 | — |
| Condensed milk | 50 | 250 | — |
| Ice cream stabilizer blend | 3 | 15 | — |
| Vanilla | 2 | 10 | — |

1. Confirm each machine model, feed system and permitted fat, solids, stabilizer and viscosity range before loading. The owner describes one machine as soft serve and the other as gravity; these categories overlap, so the second is provisionally treated as gravity-fed soft serve, with its actual product mode still to be verified. If either manual rejects this mix, use its approved commercial premix until a compatible reformulation is tested.
2. Dry-blend sugar, glucose powder, milk powder and an ice-cream-specific stabilizer/emulsifier blend. The 3 g dose is provisional: follow the actual blend label, offset any dose change with milk and recheck solids.
3. Warm milk and cream, whisk in dry blend, condensed milk and vanilla. Apply the controlled dairy processing protocol F0 before use. Reconstituted milk powder must not be treated as sterile.
4. Cool rapidly, keep covered at 1–4°C and age 4–8 hours as a texture trial if the stabilizer instructions require it. Load only sanitized equipment and cold mix; record input and output weights.
5. Freeze and dispense following the manufacturer's start-up settings. Begin testing at 30% overrun as a calculation assumption, not a machine guarantee. Never add chunks, coarse nut paste or fruit fibre to a gravity feed hopper.

Yield and process note: 1,000 g mix input; 970 g saleable frozen mass after assumed 3% loss. Estimate 6.2% fat and 35% total solids using stated ingredient compositions. The real machine's hold-up may exceed this allowance; a 1 kg pilot is a formulation trial, not necessarily a sufficient machine charge.

Illustrative ingredient cost: 2,100 FCFA per pilot batch; 2,164.9 FCFA/kg usable output. This excludes packaging and operating costs.

## FBASE2 Vanilla condensed milk popsicle mix

Pilot input 1,000 g; target usable output 970 g. Planning production batch is x5: input 5,000 g and usable output 4,850 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| UHT milk | 640 | 3,200 | — |
| Condensed milk | 180 | 900 | — |
| Cream 35% | 60 | 300 | — |
| Sugar | 70 | 350 | — |
| Skim milk powder | 45 | 225 | — |
| Ice cream stabilizer blend | 3 | 15 | — |
| Vanilla | 2 | 10 | — |

1. Dry-blend sugar, skim powder and stabilizer suitable for frozen bars; verify the blend's label dose.
2. Warm milk and cream, whisk in dry ingredients and condensed milk, then use F0 dairy processing and rapid chilling.
3. Fill cleaned moulds to their measured working volume, leaving the manufacturer-required expansion space. Insert food-safe sticks and freeze in the popsicle machine according to its manual.

4. Demould with potable-water contact controlled by the machine instructions; bag immediately and transfer to the dedicated hardening/storage freezer. Track yield by weight.

Yield and process note: 1,000 g input; 970 g usable mix after loss. Approximate 6% fat, 35% solids and 15% added sucrose before naturally occurring lactose; recheck labels. No overrun assumed.

Illustrative ingredient cost: 2,057 FCFA per pilot batch; 2,120.6 FCFA/kg usable output. This excludes packaging and operating costs.

# FBASE3 Pineapple fruit ice mix

Pilot input 1,000 g; target usable output 970 g. Planning production batch is x5: input 5,000 g and usable output 4,850 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Pineapple puree pasteurized | 500 | 2,500 | — |
| Water | 310 | 1,550 | — |
| Sugar | 140 | 700 | — |
| Glucose powder | 45 | 225 | — |
| Lemon juice | 3 | 15 | — |
| Ice cream stabilizer blend | 2 | 10 | — |

1. Use commercially pasteurized puree and potable water. Confirm the stabilizer is suitable for acidic water ices, not simply any dairy stabilizer.
2. Heat water with sugar, glucose and stabilizer according to its hydration instructions, then combine with puree using sanitized equipment. Select a supplier-approved fruit processing schedule; dairy time/temperature guidance is not a fruit-process validation.
3. Chill promptly under S1. Measure soluble solids with a refractometer; a trial target is 22–25 degrees Brix, adjusted for the actual fruit. Reweigh any formula adjustment and re-cost.
4. Fill measured moulds, insert sticks, freeze, demould hygienically, bag and harden. This formula is for moulded ices, not automatically for the soft-serve hopper.

Yield and process note: 1,000 g input; 970 g usable mix. At 12 degrees Brix puree, calculated soluble solids are roughly 24.5% before small acid/stabilizer contributions. Zero milk fat; not dairy ice cream.

Illustrative ingredient cost: 1,017 FCFA per pilot batch; 1,048.9 FCFA/kg usable output. This excludes packaging and operating costs.

# FBASE4 Mango yogurt popsicle mix

Pilot input 1,000 g; target usable output 970 g. Planning production batch is x5: input 5,000 g and usable output 4,850 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Yogurt plain | 500 | 2,500 | — |
| Mango puree pasteurized | 250 | 1,250 | — |
| Condensed milk | 120 | 600 | — |
| Sugar | 70 | 350 | — |
| Water | 58 | 290 | — |
| Ice cream stabilizer blend | 2 | 10 | — |

1. Use commercially pasteurized fruit puree and yogurt made from pasteurized milk, kept cold. Use a stabilizer intended for acidic frozen yogurt bars.
2. Hydrate stabilizer with sugar and water according to its instructions, then cool rapidly before blending into the yogurt, puree and condensed milk.
3. Blend with sanitized tools, keep at or below 5°C and fill moulds promptly. Do not claim live-culture counts or probiotic benefits.
4. Freeze, demould, bag and harden as for milk pops. This is a cold-assembled process requiring hygienic ingredients and validated local handling, not a claim of on-site pasteurization.

Yield and process note: 1,000 g input; 970 g usable mix. Provisional solids around 26% and fat around 2.7% using yogurt at 12% solids and 3.5% fat, mango at 15% solids, and the stated condensed milk assumption; actual labels control the final balance.

Illustrative ingredient cost: 1,902 FCFA per pilot batch; 1,961.1 FCFA/kg usable output. This excludes packaging and operating costs.

# FBASE5 Chocolate sundae sauce

Pilot input 955 g; target usable output 900 g. Planning production batch is x5: input 4,775 g and usable output 4,500 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Water | 400 | 2,000 | — |
| Sugar | 250 | 1,250 | — |
| Cocoa | 100 | 500 | — |
| Condensed milk | 200 | 1,000 | — |
| Salt | 2 | 10 | — |
| Vanilla | 3 | 15 | — |

1. Whisk cocoa and sugar, gradually add water and bring to a simmer while stirring.
2. Add condensed milk and salt; cook gently to 900 g finished mass without scorching, then add vanilla.
3. Cool under S1 and keep refrigerated in labelled small containers.

4. Serve 20 g cold sauce per sundae; do not return used sauce to the storage batch.

Yield and process note: Input 955 g; about 55 g evaporation. This milk-containing sauce requires refrigeration.

Illustrative ingredient cost: 1,539 FCFA per pilot batch; 1,709.8 FCFA/kg usable output. This excludes packaging and operating costs.

# FBASE6 Soft sandwich cookies

Pilot input 1,086 g; target usable output 900 g. Planning production batch is x5: input 5,430 g and usable output 4,500 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Plain flour | 500 | 2,500 | — |
| Margarine | 220 | 1,100 | — |
| Sugar | 220 | 1,100 | — |
| Egg | 100 | 500 | — |
| Cocoa | 35 | 175 | — |
| Baking powder | 4 | 20 | — |
| Salt | 4 | 20 | — |
| Vanilla | 3 | 15 | — |

1. Cream margarine and sugar just until smooth; mix in egg and vanilla.
2. Fold in sifted flour, cocoa, baking powder and salt. Chill 20 minutes, roll 3–4 mm and cut evenly.
3. Bake at 170–180°C for a trial 9–12 minutes until set but not rock-hard. Cool fully; target 900 g baked cookies.
4. Match by diameter. Use two 20 g cookies per ice cream sandwich; record breakage separately.

Yield and process note: Input 1,086 g; expected cooked usable cookies 900 g after bake/trim loss. Verify freezer bite before launch.

Illustrative ingredient cost: 1,416 FCFA per pilot batch; 1,572.9 FCFA/kg usable output. This excludes packaging and operating costs.

# FBASE7 Chocolate dipping coating

Pilot input 1,000 g; target usable output 980 g. Planning production batch is x5: input 5,000 g and usable output 4,900 g. These are trial targets; machine minimum/maximum batch capacities remain unverified.

| Ingredient or prepared component | Pilot g | Production g | Baker % |
|---|---|---|---|
| Compound chocolate coating | 950 | 4,750 | — |
| Oil | 50 | 250 | — |

1. Use a commercially supplied compound coating intended for frozen confectionery. Melt only within its manufacturer's temperature range.
2. Add the stated oil only if its instructions allow thinning; otherwise use 1,000 g coating and update cost.
3. Dip fully hardened dry milk bars briefly using clean tools. Keep moisture out of the coating bowl.
4. Return coated bars immediately to frozen storage; do not put unused dipped-product crumbs back into clean coating stock.

Yield and process note: Input 1,000 g; 20 g bowl/handling loss. Remaining coating working stock must be recorded; do not charge it repeatedly if still usable.

Illustrative ingredient cost: 4,355 FCFA per pilot batch; 4,443.9 FCFA/kg usable output. This excludes packaging and operating costs.

## F01 Vanilla soft serve cup or cone

Quality target: Smooth fine texture, clean vanilla flavour and stable shape for immediate eating.

Portion and pilot yield: 10 x 95 g; 20 g serving remainder. Costed saleable units: 10.

| Prepared component | Pilot g | Planning x5 g |
|---|---|---|
| FBASE1 Gravity soft serve vanilla trial mix | 970 | 4,850 |

1. Prepare FBASE1 under F0 and dispense using the machine manual. Weigh ten 95 g servings; a 20 g residual is charged to the batch.
2. At the illustrative 1.09 g/ml mix density and 30% overrun, 95 g occupies about 113 ml; choose a cup with adequate headroom. A cone is an alternative package, not a second product.
3. Serve immediately. Keep hopper mix cold, log temperatures and follow the specific daily dismantling/sanitation or approved heat-treatment cycle.

Compatibility and yield: Prototype for the reported soft serve and gravity machine arrangement, conditional on both manuals. Chocolate alternative per 1 kg input: replace 25 g milk with 25 g cocoa; recheck viscosity, freezing and cost. No chunk inclusions.

Troubleshooting: Icy: unstable temperature/low solids; weak swirl: formulation or machine setting; sandy: excessive milk solids or storage abuse.

Storage and release: follow F0; weigh fill/serving mass, verify cold-chain log, check stick/packaging integrity where applicable, record rejects and disclose milk, gluten, egg or nuts as the recipe requires. Cup/cone, stick/bag and carton prices are separately assumed in costing.

| Batch ingredient cost | Saleable units | Ingredients/unit | Pack/unit | Other variable/unit | Test price | Contribution/unit |
|---|---|---|---|---|---|---|
| 2,100 | 10 | 210.0 | 35 | 30 | 600 | 325.0 |

All values in FCFA. Contribution margin 54.2%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 283.0 FCFA. At 80% sell-through it falls to 256.2 FCFA per sold unit under the section 7 model.

# F02 Chocolate peanut sundae

Quality target: Clean layers, measured topping and crunch without a flooded cup.

Portion and pilot yield: 120 g soft serve + 20 g sauce + 10 g nuts. Costed saleable units: 5.

| Prepared component | Pilot g | Planning x5 g |
|---|---|---|
| FBASE1 Gravity soft serve vanilla trial mix | 600 | 3,000 |
| FBASE5 Chocolate sundae sauce | 100 | 500 |
| Peanut roasted | 50 | 250 |

1. Use FBASE1 and fully chilled FBASE5; use reputable commercially roasted peanuts, inspect and store dry.
2. Dispense 120 g ice cream into each cup, add 20 g sauce and 10 g nuts with dedicated utensils.
3. Serve immediately. Offer a nut-free recipe only where cross-contact can be controlled; removing visible nuts is insufficient.

Compatibility and yield: A service format from existing soft serve, not an extra freezer formulation. Cost uses five servings, 750 g total.

Troubleshooting: Rapid melt: warm sauce/cup; excessive cost: unmeasured nuts; stale taste: old nuts.

Storage and release: follow F0; weigh fill/serving mass, verify cold-chain log, check stick/packaging integrity where applicable, record rejects and disclose milk, gluten, egg or nuts as the recipe requires. Cup/cone, stick/bag and carton prices are separately assumed in costing.

| Batch ingredient cost | Saleable units | Ingredients/unit | Pack/unit | Other variable/unit | Test price | Contribution/unit |
|---|---|---|---|---|---|---|
| 1,560 | 5 | 312.0 | 60 | 35 | 1000 | 593.0 |

All values in FCFA. Contribution margin 59.3%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 530.6 FCFA. At 80% sell-through it falls to 491.3 FCFA per sold unit under the section 7 model.

# F03 Chocolate milkshake

Quality target: Drinkable thick consistency, no ice lumps and consistent fill.

Portion and pilot yield: 150 g soft serve + 120 g milk + 20 g sauce. Costed saleable units: 5.

| Prepared component | Pilot g | Planning x5 g |
|---|---|---|
| FBASE1 Gravity soft serve vanilla trial mix | 750 | 3,750 |
| UHT milk | 600 | 3,000 |
| FBASE5 Chocolate sundae sauce | 100 | 500 |

1. Confirm a food-safe high-speed blender is available; the mixing machine/pot alone is not evidence of blending capability.

2. Blend one 150 g ice cream portion with 120 g cold UHT milk and 20 g chilled sauce for 15–25 seconds, until just smooth.
3. Use a roughly 350 ml cup after measuring actual aeration volume. Serve immediately; wash and sanitize blender parts between production sessions and allergens.

Compatibility and yield: Conditional on blender availability. No raw egg. 1,450 g input for five portions; mass 290 g each before trivial blender residue, charged via the loss model.

Troubleshooting: Thin: overblending/warm milk; excessive sweetness: unmeasured syrup; lumps: inadequate blender.

Storage and release: follow F0; weigh fill/serving mass, verify cold-chain log, check stick/packaging integrity where applicable, record rejects and disclose milk, gluten, egg or nuts as the recipe requires. Cup/cone, stick/bag and carton prices are separately assumed in costing.

| Batch ingredient cost | Saleable units | Ingredients/unit | Pack/unit | Other variable/unit | Test price | Contribution/unit |
|---|---|---|---|---|---|---|
| 2,515 | 5 | 502.9 | 75 | 40 | 1200 | 582.1 |

All values in FCFA. Contribution margin 48.5%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 481.5 FCFA. At 80% sell-through it falls to 427.6 FCFA per sold unit under the section 7 model.

# F04 Chocolate cookie ice cream sandwich

Quality target: Biteable cookie with a firm clean filling, no large ice crystals.

Portion and pilot yield: 2 x 20 g cookies + 70 g frozen filling. Costed saleable units: 10.

| Prepared component | Pilot g | Planning x5 g |
|---|---|---|
| FBASE6 Soft sandwich cookies | 400 | 2,000 |
| FBASE1 Gravity soft serve vanilla trial mix | 700 | 3,500 |

1. Bake FBASE6, cool fully and select twenty 20 g cookies.
2. Dispense or spread FBASE1 into clean shallow moulds, portioning 70 g per filling. Harden quickly in suitable freezing equipment; storage-freezer capacity alone does not prove hardening throughput.
3. Sandwich each hardened filling between two cookies, wrap individually and return to frozen storage immediately. Pilot freezer bite and condensation resistance.

Compatibility and yield: Uses soft-serve output hardened in shallow forms; no batch freezer is assumed. Start with small production, and approve only after a recorded hardening test.

Troubleshooting: Hard cookie: overbake; crumbly assembly: unmatched diameter; icy filling: slow hardening.

Storage and release: follow F0; weigh fill/serving mass, verify cold-chain log, check stick/packaging integrity where applicable, record rejects and disclose milk, gluten, egg or nuts as the recipe requires. Cup/cone, stick/bag and carton prices are separately assumed in costing.

| Batch ingredient cost | Saleable units | Ingredients/unit | Pack/unit | Other variable/unit | Test price | Contribution/unit |
|---|---|---|---|---|---|---|
| 2,145 | 10 | 214.5 | 25 | 35 | 700 | 425.5 |

All values in FCFA. Contribution margin 60.8%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 382.6 FCFA. At 80% sell-through it falls to 356.9 FCFA per sold unit under the section 7 model.

# F05 Condensed milk popsicles

Quality target: Creamy clean release and securely centred stick.

Portion and pilot yield: Example small 50 ml mould at density 1.10 g/ml: 55 g each. Costed saleable units: 17.

| Prepared component | Pilot g | Planning x5 g |
|---|---|---|
| FBASE2 Vanilla condensed milk popsicle mix | 970 | 4,850 |

1. Prepare FBASE2 and use the popsicle machine manual for mould immersion, stick timing and freeze completion.
2. At example 50 ml working fill and 1.10 g/ml density, make 17 x 55 g bars, with 35 g remainder. At example 100 ml, make 8 x 110 g bars with 90 g remainder. Actual moulds are unconfirmed.
3. Bag each bar, include the stick cost, and move immediately to hardening/frozen storage. Use F0 storage policy.

Compatibility and yield: One product in two sizes. Larger example price 500 FCFA (test 450–650); separate cost row supplied. Densities and volumes must be measured.

Troubleshooting: Icy: low solids/slow freezing; too soft: excessive sugar; pulled-out stick: incomplete freezing.

Storage and release: follow F0; weigh fill/serving mass, verify cold-chain log, check stick/packaging integrity where applicable, record rejects and disclose milk, gluten, egg or nuts as the recipe requires. Cup/cone, stick/bag and carton prices are separately assumed in costing.

| Batch ingredient cost | Saleable units | Ingredients/unit | Pack/unit | Other variable/unit | Test price | Contribution/unit |
|---|---|---|---|---|---|---|
| 2,057 | 17 | 121.0 | 20 | 20 | 300 | 139.0 |

All values in FCFA. Contribution margin 46.3%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 114.8 FCFA. At 80% sell-through it falls to 98.8 FCFA per sold unit under the section 7 model.

# F06 Pineapple fruit ice pops

Quality target: Distinct fruit flavour and biteable ice texture, not plain coloured sugar water.

Portion and pilot yield: Example small 50 ml x 1.10 g/ml = 55 g. Costed saleable units: 17.

| Prepared component | Pilot g | Planning x5 g |
|---|---|---|
| FBASE3 Pineapple fruit ice mix | 970 | 4,850 |

1. Make FBASE3; measure fruit solids and density before filling.
2. Use the same illustrative 17 small or 8 large bar yield calculation as F05 only if measured density is 1.10 g/ml. Freeze, demould and bag under the machine procedure.
3. Offer the large 100 ml example at 400 FCFA (test 350–500). Keep a separate dairy-contact workflow if marketing as dairy-free.

Compatibility and yield: One product in two sizes. Do not feed this fruit formula into gravity soft serve unless its manual expressly permits it.

Troubleshooting: Rock-hard: too little sugar/solids; sticky soft: excessive Brix; weak fruit: diluted puree.

Storage and release: follow F0; weigh fill/serving mass, verify cold-chain log, check stick/packaging integrity where applicable, record rejects and disclose milk, gluten, egg or nuts as the recipe requires. Cup/cone, stick/bag and carton prices are separately assumed in costing.

| Batch ingredient cost | Saleable units | Ingredients/unit | Pack/unit | Other variable/unit | Test price | Contribution/unit |
|---|---|---|---|---|---|---|
| 1,017 | 17 | 59.9 | 20 | 18 | 250 | 152.2 |

All values in FCFA. Contribution margin 60.9%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 140.2 FCFA. At 80% sell-through it falls to 127.7 FCFA per sold unit under the section 7 model.

# F07 Mango yogurt frozen bar

Quality target: Balanced mango acidity, smooth bite and firm shape.

Portion and pilot yield: Example 70 g bars: 12 units; 130 g remainder. Costed saleable units: 12.

| Prepared component | Pilot g | Planning x5 g |
|---|---|---|
| FBASE4 Mango yogurt popsicle mix | 970 | 4,850 |

1. Make FBASE4 with hygienic ready-to-eat ingredients. Portion 70 g each into a mould with sufficient verified working volume.
2. Freeze promptly, demould hygienically and wrap. Charge the 130 g remainder to the twelve bars until a measured mould/batch combination reduces it.
3. Run a sourness and freezer-bite test; declare milk and do not imply health or live-culture benefits.

Compatibility and yield: Moulded yogurt dessert, not a gravity-machine frozen-yogurt recipe. Approximate 64 ml at 1.10 g/ml density, subject to measurement.

Troubleshooting: Curdled/grainy: acid imbalance; icy: low solids; weak mango: poor puree.

Storage and release: follow F0; weigh fill/serving mass, verify cold-chain log, check stick/packaging integrity where applicable, record rejects and disclose milk, gluten, egg or nuts as the recipe requires. Cup/cone, stick/bag and carton prices are separately assumed in costing.

| Batch ingredient cost | Saleable units | Ingredients/unit | Pack/unit | Other variable/unit | Test price | Contribution/unit |
|---|---|---|---|---|---|---|
| 1,902 | 12 | 158.5 | 25 | 25 | 450 | 241.5 |

All values in FCFA. Contribution margin 53.7%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 209.8 FCFA. At 80% sell-through it falls to 189.3 FCFA per sold unit under the section 7 model.

## F08 Chocolate dipped peanut milk bar

Quality target: Thin crisp coating, creamy core and clean wrapper.

Portion and pilot yield: 66 g milk bar + 18 g coating allocation + 5 g peanut. Costed saleable units: 10.

| Prepared component | Pilot g | Planning x5 g |
|---|---|---|
| FBASE2 Vanilla condensed milk popsicle mix | 660 | 3,300 |
| FBASE7 Chocolate dipping coating | 180 | 900 |
| Peanut roasted | 50 | 250 |

1. Freeze ten 66 g FBASE2 cores using an appropriately sized mould and harden fully.
2. Use FBASE7 following coating supplier temperature instructions; dip dry cold cores quickly. Target 15 g adhering chocolate per bar; the 18 g allocation includes 3 g drips/handling.
3. Sprinkle 5 g roasted peanut on each before coating sets, then wrap and freeze immediately. Keep nuts segregated.

Compatibility and yield: Approximate sale weight 86 g; 30 g coating loss across ten bars is charged. This is an added-value process and allergen burden, not simply a new flavour.

Troubleshooting: Coating falls off: surface frost; thick shell: wrong viscosity; cracking: excessive coating.

Storage and release: follow F0; weigh fill/serving mass, verify cold-chain log, check stick/packaging integrity where applicable, record rejects and disclose milk, gluten, egg or nuts as the recipe requires. Cup/cone, stick/bag and carton prices are separately assumed in costing.

| Batch ingredient cost | Saleable units | Ingredients/unit | Pack/unit | Other variable/unit | Test price | Contribution/unit |
|---|---|---|---|---|---|---|
| 2,290 | 10 | 228.9 | 30 | 35 | 600 | 306.1 |

All values in FCFA. Contribution margin 51.0%; this is before payroll, rent and other fixed overhead. 20% ingredient-price increase reduces contribution to 260.3 FCFA. At 80% sell-through it falls to 232.6 FCFA per sold unit under the section 7 model.

# 7 Costing pricing and operating profitability

Revision note: the original product tables remain baseline recipe estimates. For the added preparation-ahead route, section 5A provides extra reheating-loss allowances and new A01–A04 costs; update cold handling, transport, energy and labour before relying on these margins.

The following are transparent planning estimates, not quotations, current local prices or profit promises. Replace every ingredient and packaging input with a dated delivered supplier quote before using the plan to purchase or price stock. Whole milk powder and skim milk powder are different inputs. Edible-weight conversion matters: a 1,000 FCFA/kg gross ingredient with 20% trim loss costs 1,250 FCFA/kg usable, before labour.

## Editable ingredient price assumptions

| Ingredient | FCFA/kg | Ingredient | FCFA/kg |
|---|---|---|---|
| Baking powder | 4,000 | Lettuce trimmed | 1,800 |
| Baking soda | 2,500 | Mango puree pasteurized | 1,800 |
| Banana peeled | 800 | Margarine | 1,800 |
| Beef boneless | 4,500 | Mayonnaise pasteurized | 2,500 |
| Beef mince | 4,500 | Mozzarella | 6,500 |
| Black pepper | 9,000 | Mustard | 3,000 |
| Bread flour | 650 | Nutmeg | 10,000 |
| Breadcrumbs | 1,000 | Oil | 1,600 |
| Butter | 7,000 | Onion trimmed | 800 |
| Cabbage trimmed | 600 | Oregano | 12,000 |
| Canned fish drained | 4,000 | Paprika | 8,000 |
| Carrot trimmed | 900 | Peanut roasted | 1,800 |
| Cheese slices | 6,500 | Pepper trimmed | 1,200 |
| Chicken boneless | 4,000 | Pineapple puree pasteurized | 1,500 |
| Chili trimmed | 2,000 | Plain flour | 650 |
| Cinnamon | 8,000 | Potato peeled | 900 |
| Cocoa | 7,000 | Salt | 400 |
| Compound chocolate coating | 4,500 | Sausage meat | 4,000 |
| Condensed milk | 3,000 | Sesame | 3,000 |
| Coriander | 8,000 | Skim milk powder | 5,000 |
| Corn starch | 1,800 | Sugar | 800 |
| Cream 35% | 6,500 | Sweet potato flour | 1,000 |
| Cucumber trimmed | 800 | Tahini | 7,000 |
| Cumin | 8,000 | Thyme | 10,000 |
| Curry powder | 6,000 | Tomato trimmed | 800 |

| Ingredient | FCFA/kg | Ingredient | FCFA/kg |
|---|---|---|---|
| Egg | 2,200 | UHT milk | 1,200 |
| Garlic peeled | 3,000 | Vanilla | 12,000 |
| Ginger peeled | 2,000 | Vinegar 5% | 900 |
| Glucose powder | 2,500 | Water | 5 |
| Ice cream stabilizer blend | 18,000 | Whole milk powder | 4,500 |
| Icing sugar | 1,400 | Yeast | 6,000 |
| Ketchup | 1,800 | Yogurt plain | 2,000 |
| Lemon juice | 1,800 | | |

Egg example: a 50 g edible egg at the assumed 2,200 FCFA/kg costs 110 FCFA. This is a price assumption, not a local market observation. Quotes for milk sold by litre must be converted with measured density; do not silently treat oil litres as kilograms. Packaging assumptions include the named container, with a food-safe stick for popsicles and a spoon/straw where relevant.

# How the model works

1. Batch ingredient cost = sum of ingredient grams × FCFA/kg ÷ 1,000. For a prepared component, recursively include all its raw ingredients and divide by its usable output, not raw input mass.
2. Ingredient cost per saleable unit = costed batch / stated saleable units. Explicit pastry/dough QC allowances are charged to the product. Full unused filling batches are not automatically charged twice: each assembly card charges only its required usable share, so unused safe stock must be recorded and eventually sold or written off.
3. Variable cost per unit = ingredient cost + packaging + other variable allowance. Contribution = selling price − variable cost. Contribution margin = contribution / selling price. Contribution does not pay the owner until overhead and other obligations are met.
4. Other-variable allowances are placeholders for production energy/diesel, incremental cleaning/water and variable handling/delivery. Provisionally allocate 60% to energy/diesel, 20% to incremental cleaning/water and 20% to variable handling/delivery, then replace with actual logs. Paid shift wages and base route transport are in overhead below; do not charge the same expense in both places.
5. Sensitivity at 20% ingredient inflation: contribution falls by 0.20 × ingredient cost per unit. At 80% sell-through, assume all planned variable cost was incurred, including packaging: effective contribution per sold unit = price − full planned variable cost / 0.80. This conservatively charges unsold units to those sold.
6. All prices and calculations are gross cash planning values without a tax determination. If an applicable tax must be remitted from the displayed selling price, remove it from revenue before computing contribution. Obtain the business-specific treatment locally; no Cameroon tax status is assumed.

# Core product contribution table

| Product | Ingredients/unit | Pack + other | Test price | Contribution | Margin |
|---|---|---|---|---|---|
| P01 Soft family bread | 278.4 | 30 + 55 | 650 | 286.6 | 44.09% |
| P02 Condensed milk breakfast buns | 92.6 | 10 + 20 | 250 | 127.4 | 50.95% |
| P03 Nkambe sweet potato milk loaf | 436.2 | 35 + 60 | 800 | 268.8 | 33.6% |
| P04 Sugar ring doughnuts | 73.3 | 10 + 25 | 250 | 141.7 | 56.67% |
| P05 Puff puff | 18.9 | 3 + 8 | 75 | 45.1 | 60.13% |
| P06 Cameroon style fish rolls | 139.4 | 10 + 30 | 300 | 120.6 | 40.19% |
| P07 Beef meat pies | 190.5 | 20 + 35 | 500 | 254.5 | 50.9% |
| P08 Sausage rolls | 205.3 | 15 + 35 | 450 | 194.7 | 43.26% |
| P09 Creamy chicken pies | 229.6 | 20 + 40 | 650 | 360.4 | 55.44% |
| P10 Whole egg dough rolls | 197.2 | 15 + 35 | 500 | 252.8 | 50.56% |
| P11 Banana condensed milk cake slices | 153.6 | 20 + 25 | 400 | 201.4 | 50.36% |
| P12 Personal chicken pizza | 780.1 | 90 + 80 | 1,800 | 849.9 | 47.22% |
| P13 Complete chicken shawarma | 957.5 | 80 + 60 | 2,000 | 902.5 | 45.12% |
| P14 Complete beef shawarma | 1,059.7 | 80 + 65 | 2,400 | 1,195.3 | 49.8% |
| P15 Crispy chicken burger | 660.2 | 70 + 65 | 1,800 | 1,004.8 | 55.82% |
| P16 Beef cheeseburger | 780.7 | 70 + 60 | 1,900 | 989.3 | 52.07% |
| P17 Crispy chicken and fries box | 1,272.3 | 100 + 100 | 2,500 | 1,027.7 | 41.11% |
| P18 Chicken vegetable samosas | 106.3 | 10 + 25 | 300 | 158.7 | 52.9% |
| P19 Loaded chicken shawarma fries | 895.9 | 90 + 60 | 1,800 | 754.1 | 41.89% |
| P20 Condensed milk soaked cake portions | 334.2 | 45 + 35 | 650 | 235.8 | 36.28% |

# Frozen dessert contribution table

| Product | Batch cost / units | Ingredients/unit | Pack + other | Test price | Contribution |
|---|---|---|---|---|---|
| F01 Vanilla soft serve cup or cone | 2,100 / 10 | 210.0 | 35 + 30 | 600 | 325.0 |
| F02 Chocolate peanut sundae | 1,560 / 5 | 312.0 | 60 + 35 | 1,000 | 593.0 |
| F03 Chocolate milkshake | 2,515 / 5 | 502.9 | 75 + 40 | 1,200 | 582.1 |
| F04 Chocolate cookie ice cream sandwich | 2,145 / 10 | 214.5 | 25 + 35 | 700 | 425.5 |
| F05 Condensed milk popsicles | 2,057 / 17 | 121.0 | 20 + 20 | 300 | 139.0 |
| F06 Pineapple fruit ice pops | 1,017 / 17 | 59.9 | 20 + 18 | 250 | 152.2 |

| Product | Batch cost / units | Ingredients/unit | Pack + other | Test price | Contribution |
|---|---|---|---|---|---|
| F07 Mango yogurt frozen bar | 1,902 / 12 | 158.5 | 25 + 25 | 450 | 241.5 |
| F08 Chocolate dipped peanut milk bar | 2,290 / 10 | 228.9 | 30 + 35 | 600 | 306.1 |
| F05L Condensed milk popsicles large example | 2,057 / 8 | 257.1 | 25 + 25 | 500 | 192.9 |
| F06L Pineapple fruit ice pops large example | 1,017 / 8 | 127.2 | 25 + 22 | 400 | 225.8 |

The large pop is twice the small fill volume, but the selling price need not be twice the small price because one stick/bag and part of the service effort are shared per unit. The lower large-milk-pop margin here also reflects conservative pilot remainders. Do not infer a steady production margin from a small batch that leaves most of a cavity unused.

## Recipe and energy cost reality checks

Bread uses approximately 278 FCFA ingredients per loaf at the assumed flour price, before packaging and energy. Its 650 FCFA test price is not automatically affordable locally; test against actual competing loaf weights. P03 sweet potato bread and P20 soaked cake have lower example contribution margins than several snacks. Their roles are differentiation and basket value, not guaranteed volume or margin leadership.

The family loaf's 55 FCFA other-variable allowance assigns 33 FCFA to energy/diesel provisionally. If an oven load produces 48 loaves, the allocated energy budget is 1,584 FCFA per load. Compare that with measured diesel litres × delivered diesel price plus electricity/battery-related costs. If the actual energy cost per loaf is higher, replace the allowance; do not declare the loaf profitable from ingredient cost alone.

| Example daily fixed/standing cost item | FCFA/day assumption | Measurement needed |
|---|---|---|
| Paid production and sales shifts including owner replacement labour | 18000 | Actual staffing hours, wages and employer costs |
| Rent and standing facility expenses | 3500 | Both premises monthly costs / actual trading days |
| Base transport route | 2500 | Vehicle/fuel/driver allocation; incremental deliveries above excluded |
| Standing refrigeration and backup power costs | 2500 | Metered overnight/standby power and backup cost |
| Equipment and solar/battery depreciation or replacement reserve | 2500 | Asset price, expected life and maintenance plan |
| Routine deep cleaning and pest control | 1000 | Base cleaning labour/consumables not already variable |
| Administration and communications | 1000 | Actual fixed expenses |
| Total illustrative daily overhead | 31000 | Excludes financing, income tax and unexpected losses |

# Daily basket illustration and break even

| Product | Units prepared | Price | Revenue if all sell | Planned variable cost |
|---|---|---|---|---|
| P01 Soft family bread | 30 | 650 | 19,500 | 10,902 |
| P02 Condensed milk breakfast buns | 50 | 250 | 12,500 | 6,132 |
| P04 Sugar ring doughnuts | 40 | 250 | 10,000 | 4,333 |
| P05 Puff puff | 60 | 75 | 4,500 | 1,794 |
| P06 Cameroon style fish rolls | 30 | 300 | 9,000 | 5,383 |
| P07 Beef meat pies | 24 | 500 | 12,000 | 5,892 |
| P13 Complete chicken shawarma | 12 | 2000 | 24,000 | 13,170 |
| F01 Vanilla soft serve cup or cone | 20 | 600 | 12,000 | 5,500 |
| F05 Condensed milk popsicles | 17 | 300 | 5,100 | 2,737 |

This illustration prepares 283 units in the stated mix, not a demand forecast. Revenue at 100% sale is 108,600 FCFA; planned variable cost 55,844; contribution 52,756. After 31,000 FCFA illustrative overhead, operating surplus is 21,756 FCFA before financing, income tax and exceptional losses. Filling and mix start-up stock can require more cash than these consumed-portion costs.

| Scenario | Revenue | Variable cost incurred | Overhead | Operating result |
|---|---|---|---|---|
| 100% sold, base prices | 108,600 | 55,844 | 31,000 | 21,756 |
| 80% sold, all production costs incurred | 86,880 | 55,844 | 31,000 | 36 |
| 60% sold, all production costs incurred | 65,160 | 55,844 | 31,000 | -21,684 |
| 100% sold, ingredient costs +20% | 108,600 | 64,555 | 31,000 | 13,045 |
| 80% sold and ingredient costs +20% | 86,880 | 64,555 | 31,000 | -8,675 |

At full sell-through and unchanged mix, average contribution is 186.4 FCFA/unit; break-even volume is about 167 equivalent units/day, or about 63,814 FCFA sales. If the entire illustrated basket is already produced, its cash break-even sell-through is 80.0%. These are different questions. Changing the product mix, waste rate or staffing changes both thresholds.

# Price ladder and simple combinations

Use visible single-item prices and portion descriptions. Entry: 75 FCFA puff puff, 250 FCFA bun/doughnut, 250–300 FCFA small pop. Everyday: 300 FCFA fish roll, 500 FCFA beef pie, 650 FCFA family bread. Treat: 600 FCFA soft serve or 650 FCFA milk cake. Meal: 1,800–2,400 FCFA pizza, burgers and wraps. These are test points; an expensive meal should not displace the affordable traffic range.

Two milk buns: example bundle price 450 FCFA; ingredient/pack/other variable cost 245.3; contribution 204.7 before overhead. Use only when it adds purchases rather than discounting an order the customer would have made anyway.

Beef pie and small milk pop: example bundle price 750 FCFA; ingredient/pack/other variable cost 406.5; contribution 343.5 before overhead. Use only when it adds purchases rather than discounting an order the customer would have made anyway.

Chicken shawarma and small fruit pop: example bundle price 2,200 FCFA; ingredient/pack/other variable cost 1,195.4; contribution 1,004.6 before overhead. Use only when it adds purchases rather than discounting an order the customer would have made anyway.

For the first week, capture actual supplier cost, produced units, rejects, samples, sales, discounts, unsold units and customer comments daily. Update the CSV inputs and re-run the calculations before changing the menu. The supplied CSVs are editable data tables, not a live automatically recalculating spreadsheet; the accompanying Python builder regenerates the report from changed inputs.

# 8 Two site production staffing and energy plan

## Site responsibilities and transport

Proposed allocation, subject to the actual installed locations: production handles dough, pastry, bread, buns, cakes, cooked fillings and validated frozen-mix preparation. Sales handles shawarma final cooking, burgers, fries, soft-serve dispensing and final assembly. Popsicle production belongs at whichever site has suitable sanitation, rapid hardening and enough power headroom; its unknown rating prevents a firm allocation.

Use separate clean food-safe crates for cooled bread, insulated chilled containers for perishable fillings/mix, and insulated frozen transport for bars. Raw meat travels sealed and segregated below ready-to-eat food. Record dispatch and receipt time, batch, quantity and product temperature. Chilled food must meet S1; hot transport must maintain the house hot standard; frozen products should remain frozen at the operating target. Route duration and container capacity require a measured trial.

The receiver checks seal condition, label, batch identity and temperature before accepting. Reject unknown histories and investigate failed deliveries. Do not put newly hot food alongside chilled dairy, or close hot bread into airtight plastic. Return empty crates through a cleaning route, separate from new food.

## Illustrative daily sequence

| Window | Production site | Sales site | Main constraint |
|---|---|---|---|
| 05:30–06:00 | Check water, cold logs, battery state, diesel and scales | Receive opening stock/check cabinets when staff arrive | Early electrical work may be battery-fed |
| 06:00–07:00 | Mix and proof first bread/buns; stagger spiral and cooker | Set chilled garnish station; sanitize machine | Do not assume sunrise PV covers mixing |
| 07:00–08:30 | Bake first wave; cooling racks; start pie filling only with cooling space | Sell prior prepared fresh-delivery stock; small fry batches | Separate hot bread from chilled transport |
| 08:30–10:30 | Pie and snack batches; second bread wave if ordered | Morning snack sales; log demand and run safe replenishment | Fryer recovery and hand assembly |

| Window | Production site | Sales site | Main constraint |
|---|---|---|---|
| 10:30–12:00 | Dispatch controlled lunch components; prepare limited frozen stock | Prepare limited shawarma cone/pan portions | No oversized cone based only on oven capacity |
| 12:00–14:00 | Smaller second batch only against sell-through | Lunch service; stagger fryer, hot plate and machine loads | 15 kW sales inverter and compressor surges |
| 14:00–16:00 | Mix/dessert prep during measured solar surplus if available | Frozen dessert and snack peak hypothesis to test | Hardening capacity and battery recharge |
| 16:00–18:30 | Targeted evening bread dispatch; finish early enough to cool | Evening wraps and frozen service | Use forecast, not automatic full batches |
| Close | Waste log, cleaning, cold checks and next-day quantities | Dispose per policy, machine cleaning and safe overnight cold loads | Protect cold storage battery reserve |

# Staffing and work design

Opening working model: one trained lead baker; one prep/finishing assistant with cold-chain responsibility; one sales cook who owns raw-meat controls; one service/cashier person. A relief/delivery role may be needed during peaks and cleaning. These are roles, not proof that four people can cover all hours or all 28 products. Observe labour minutes for the limited menu before adding items. The overhead allowance must be replaced by actual paid hours, including owner work.

Label each weighed kit before mixing. Maintain a batch sheet for dough, filling and finished units. Share B07 pastry across three pies/rolls; B18 chicken across wraps, pizza and loaded fries; B20/B28 sauces across defined products; B25 buns across burgers; FBASE1 across cups, sundaes, shakes and sandwiches. Sharing reduces ingredient variety but never permits unsafe carryover or cross-contamination.

# Demand and replenishment method

Start with customer commitments and small trials. During the first three days, prepare at most one economical equipment-minimum batch per launch product, split between opening and later waves where fermentation allows. Record sales in hourly blocks. Days 4–14: forecast from comparable trading days plus paid orders; prepare about 60% of the forecast first, then release a second batch only when remaining sales and production lead time justify it. These percentages are management trial rules, not demand evidence.

Use a practical reorder trigger: expected sales during production plus delivery lead time, plus a small measured buffer, minus saleable stock already present. Do not replenish perishable stock close to closing simply to make a counter look full. Record unsold stock by cause: forecast error, poor quality, temperature failure, transport damage, portion rejects or customer preference.

# Electrical loads and daily energy

Power in kW is simultaneous demand. Energy in kWh is power × operating hours × duty fraction. Thirty kWh nominal battery energy does not mean 30 kW continuous output. Two separate battery banks cannot share load unless specifically interconnected. The following site allocation, hours and duty fractions are illustrations, not measurements. It assumes one appliance at each stated rating.

# Production illustrative electrical energy

| Load | Rated/assumed kW | Hours on | Duty fraction | kWh/day |
|---|---|---|---|---|
| Oven auxiliaries | 3 | 4 | 1 | 12.00 |
| Spiral mixer | 6.5 | 1 | 1 | 6.50 |
| Moulder | 1 | 0.5 | 1 | 0.50 |
| Proofer | 1 | 4 | 0.5 | 2.00 |
| Mixing machine | 2 | 0.5 | 1 | 1.00 |
| Induction cooker | 5 | 2 | 0.6 | 6.00 |
| Slicer unknown assumed | 0.5 | 0.5 | 1 | 0.25 |
| Cold storage aggregate assumed | 1.2 | 24 | 0.5 | 14.40 |
| Lights and small loads assumed | 0.3 | 10 | 1 | 3.00 |
| Total electrical energy | | | | 45.65 |

# Sales illustrative electrical energy

| Load | Rated/assumed kW | Hours on | Duty fraction | kWh/day |
|---|---|---|---|---|
| Ice cream machine A | 4 | 8 | 0.45 | 14.40 |
| Ice cream machine B | 1.5 | 8 | 0.45 | 5.40 |
| Shawarma | 3.5 | 4 | 0.7 | 9.80 |
| One induction fryer | 5 | 4 | 0.6 | 12.00 |
| Hot plate/cooker assumed allocation | 5 | 2 | 0.6 | 6.00 |
| Cold storage aggregate assumed | 1 | 24 | 0.5 | 12.00 |
| Lights and small loads assumed | 0.3 | 10 | 1 | 3.00 |
| Total electrical energy | | | | 62.60 |

The popsicle machine, professional pot heating, ventilation, pumps, water heating and any extra fryers/proofers are excluded because their ratings or quantities are unknown. They must be added before this is an operational load budget. Oven diesel is excluded from electrical kWh and must be measured separately. Startup and cleaning cycles can exceed the assumed running duty fractions.

# Simultaneous load examples

| Scenario | Illustrative total | Implication |
|---|---|---|
| Production oven + spiral + proofer + cold + small loads | 3 + 6.5 + 1 + 1.2 + 0.3 = 12.0 kW | Below 20 kW nominal before surges/unknown loads |

| Scenario | Illustrative total | Implication |
|---|---|---|
| Add production cooker + mixer + moulder + assumed slicer | 12 + 5 + 2 + 1 + 0.5 = 20.5 kW | Exceeds 20 kW; schedule these apart |
| Sales both ice machines + shawarma + fryer + cold + small loads | 4 + 1.5 + 3.5 + 5 + 1 + 0.3 = 15.3 kW | Exceeds 15 kW even before startup surge |
| Add sales hot plate/cooker | 15.3 + 5 = 20.3 kW | Not an acceptable all-on schedule |
| Sales A + shawarma + cold + lights | 4 + 3.5 + 1 + 0.3 = 8.8 kW | One possible low-load service block, still verify phase/surge |
| Sales A + fryer + cold + lights | 4 + 5 + 1 + 0.3 = 10.3 kW | Another block; do not add uncounted hot plate/popsicle load |

Turning a machine off is not a food-safety control. Maintain refrigeration and any active cone's continuous cooking requirements. If the safe process requires simultaneous loads beyond the inverter's verified ability, reduce the menu or use a professionally designed supply/load-control change; do not alternate a partially cooked cone into unsafe warm waiting. Check single-phase/three-phase limits and battery BMS discharge capability with an electrician before adopting any schedule.

## Battery runtime illustration

At each site, assume 30 kWh nominal × 80% usable fraction × 90% conversion efficiency = 21.6 kWh delivered AC energy. These 80% and 90% figures are assumptions, not the battery specification. Without solar/grid input, that supports a constant 1 kW load for about 21.6 hours, 2 kW for 10.8 hours, 5 kW for 4.3 hours, or 10 kW for 2.2 hours, subject to discharge power limits. Actual cycling refrigerator loads should be measured over a full day.

A twelve-hour overnight critical load averaging 1 kW needs 12 kWh delivered AC, equivalent to about 16.7 kWh nominal under the same assumptions. Set the evening reserve using measured essential refrigeration plus margin and forecast morning demand. Do not consume the cold-chain reserve on optional fryers or a second frozen-dessert machine.

PV energy cannot be estimated from the inverter size. Use measured site generation or actual installed panel kWp × site-specific equivalent sun hours × system performance ratio. The illustrative daily electrical totals above are about 45.6 kWh at production and 62.6 kWh at sales before omitted loads. Neither is covered merely because a 30 kWh battery exists; daily replenishment and evening reserve matter.

# 9 Launch validation and daily controls

## Phased launch

| Phase | Range and work | Advance only when |
|---|---|---|
| Preparation week | Supplier quotes; oven tray map; five core recipe pilots; probes/scales; machine manuals; transport and power tests | Measured yield, cooking/cooling logs and electrical plan are workable |
| Week 1 | P01, P02, P04, P05, P06, P07; small paid P03 taste test | Repeatable portions, positive contribution after waste, staff can replenish safely |

| Phase | Range and work | Advance only when |
|---|---|---|
| Week 2 | Add P13 plus either F01 or F05 after its dedicated commissioning | Recorded service drill, cold chain and machine cleaning pass |
| Weeks 3–4 | Add selected P09/P11/P18 and a second frozen format; test standard vs large pop demand | Existing range is stable and added work does not create queues/waste |
| Later development | P12, P14–P17, P19–P20 and frozen sandwiches/coated bars as justified | Measured incremental demand and margin justify equipment, labour and storage time |

# Pilot approval record

| Test | Record | Proposed acceptance rule |
|---|---|---|
| Formula | Ingredient brand, exact mass, water, dough temperature | No unrecorded substitutions; all ingredients traceable |
| Yield | Raw input, usable component mass, units, cooked mass, rejects | Three successive batches with explained variance; aim within 5% of adopted target |
| Portion | Ten random pieces per batch | Aim within ±3% of target average; no underweight labelled product |
| Sensory | Blind taste from intended customers, not only staff | At least 20 paid/structured test responses per candidate; agree threshold before testing |
| Holding quality | Check at service, 1 hour and 2 hours; longer only under safe control | No significant loss of intended texture at the planned sale window |
| Safety | Cooking, cooling, storage and transport records | All adopted critical limits met; any failure blocks sale |
| Cost | Actual delivered ingredients, oil and packaging | Positive contribution with measured waste; overhead coverage assessed |
| Service | Timed assembly and queue length | Meet practical service target without bypassing hygiene |
| Market | Units produced, sold, samples and repeats | Track 7–14 days before claiming repeat demand |
| Frozen bars | Mould ml, fill g, density, hardening time, seal/stick integrity | No unverified volume claim; stable texture and intact packaging |

The numerical business acceptance targets are proposed management rules, not regulatory standards. Adjust after baseline measurements, but do not lower food-safety limits simply to pass a commercial test. Taste tests do not establish microbial safety. Quantities in this report are recipes for controlled trials until the team demonstrates repeatability.

# Allergens and cleaning

Create a matrix from actual supplier labels. Wheat is present in breads, pastry, doughnuts, wrappers, buns, cake and cookies; milk in enriched doughs, creamy fillings, cheese, many sauces and frozen dairy; egg in egg wash, cakes, egg rolls and many commercial mayonnaise products; fish in P06; sesame in B35/tahini and optional bun topping; peanuts in F02 and F08; mustard in B28. Margarine, sausage meat, coatings, chocolate and stabilizer blends can introduce soy, milk or other allergens. Supplier labels override this preliminary list.

Use dedicated fish frying equipment or a documented separation plan; shared oil can transfer allergens and flavours to doughnuts and fries. Never label a product allergen-free merely because its formula omits the allergen. Separate raw meat tools, cooked-food tools, bakery utensils and frozen-dessert tools. Clean and sanitize according to the chemical label; do not mix chemicals or invent sanitizer concentrations.

## Power interruption and cold-chain response

1. Record outage start, cold cabinet temperatures and battery state immediately. Keep doors shut and reserve power for food-critical refrigeration.
2. Pause new high-risk preparation if safe cooling or storage cannot be maintained. Use temperature-verified backup storage/transport if available.
3. Record product temperatures and duration, not just the freezer display. Quarantine food with an unknown history; do not assume refreezing restores safety.
4. Restart machines only after safe equipment checks and the manufacturer's sanitation/recovery procedure. Discard questionable hopper mix and record the loss.
5. Review the event, reserve setting and load schedule before resuming the full menu.

## Daily opening and closing sheet

| Opening | During service | Closing |
|---|---|---|
| Probe/scale check; chilled/frozen logs; potable water; handwash; battery reserve; diesel; clean equipment | Batch IDs; cook/cool checks; hourly stock; portions; machine temperatures; delivery acceptance; actual load peaks | Unsold stock by cause; safe disposal; cash/sales reconciliation; machine strip-clean; freezer log; reserve confirmation; next-day batch plan |

Keep purchase invoices, ingredient lot numbers, preparation dates, temperatures and sales batch links. Avoid statements such as "safe for three days" or "high margin" on staff sheets unless the process and numbers support them. Assign one person to authorize formula changes and update the weighed batch sheet; improvisation should not silently become the standard.

# 10 Sources outstanding information and next actions

Research accessed 3 October 2026. External references below support the specific technical/context claims, not the entire business forecast or recipe library. Recipes, price assumptions, schedules and commercial choices are original planning proposals. No manufacturer model for the owner's machines was supplied; cited machine pages illustrate terminology and the need for model-specific instructions, not compatibility certification.

Local context: CLGF, 15 September 2025, Nkambe sweet potato project. Supports a local processing initiative; does not establish current supplier capacity or demand.

Cooking benchmark: USDA FSIS minimum internal temperature chart. External reference for poultry, ground meat and reheating. Not claimed as Cameroon law.

Cooling and chilled holding: FDA Food Code 2022 print edition, sections 3-501.14–16. Cited edition supports the chosen numerical model benchmarks; not a claim that it is the latest local rule.

Hot holding: UK food hygiene business guidance. Supports 63°C house benchmark; external jurisdiction.

Shawarma cone handling: Brighton and Hove council doner safety guidance. Search-index text supported cooked-layer shaving and conservative disposal at close; full page retrieval was unavailable.

Condensed milk label example: Nestlé Professional Australia sweetened condensed milk. Shows total sugar/fat and milk allergen for that specific product; inspect the purchased local product.

Frozen dairy processing: University of Guelph Dairy Science and Technology, Pasteurization. Technical time/temperature examples; proper equipment and validated processing remain necessary.

Overrun: University of Guelph Ice Cream Technology, Overrun calculations. Mass/volume calculation method; actual overrun and density must be measured.

Machine terminology: Carpigiani soft-serve product categories. Confirms gravity-fed soft serve is a category, not a separate frozen product.

Model-dependent hopper control: Taylor C161 product page. Illustrates model-specific separate hopper refrigeration; owner machine not identified.

# Outstanding information

| Priority | Information | Why it changes the plan |
|---|---|---|
| Before equipment loading | Machine model plates and manuals; minimum/maximum batches; hot plate/blender confirmation | Mix compatibility, safe startup, realistic throughput and missing equipment |
| Before energy commitment | PV panel kWp, actual generation, inverter phase/surge/output specs, battery BMS limits, full appliance inventory | 20/15 kW inverter labels and 30 kWh batteries do not establish autonomy |
| Before final frozen pricing | Measured working ml of both popsicle sizes, cavities and cycle time | Only the 2:1 ratio is known; all ml-based costs are examples |
| Before menu pricing | Delivered ingredient, edible-yield, packaging, diesel and labour quotations | Replace the complete illustrative cost basis |
| Before scaling | Oven diesel/hour, tray loading, mixer limits, cooling/hardening logs and transport test | Safely supported batch size and true unit cost |
| Before commercial claims | Local licensing/inspection requirements and qualified food-process review | No local legal compliance or shelf-life certification has been made |

Immediate owner actions: record the two ice-cream model numbers; measure one small and one large mould with water; obtain three quotations for flour, oil, dairy, chicken, beef and packaging; test P01, P02 and P04 in small batches; log one full day of site electrical demand and refrigeration cycling; then update the price inputs and approve the first six-product menu.